

This lecture note was done by Choongrak Kim(Professor, Department of Statistics, Pusan National University), based on “Regression Analysis” (Authors : Choongrak Kim and Geunseok Kang ; Publisher : Gyowoo Sa, 2nd edition) which is written in Korean. In fact, this note is an English version of the text.

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Chapter 5. Selection of Regression Models

5.1 Model Selection

5.1.1 Model Selection

- Two goals of regression analysis
 - (i) fit the relation between the response and the covariates
 - (ii) prediction based on the fitted model
- Model selection criterion : prediction error

5.1.2 Prediction Error

- Data
 - (i) *training data* : data used in model fitting
 - (ii) *test data* : data used in assessing the goodness-of-fit of the fitted model
- Prediction error

Let $\mathbf{X} = (X_1, \dots, X_p)$ be covariates, Y be the response, and f be the regression function to be estimated. If L is a loss function, then

$$f = \underset{g}{\operatorname{argmin}} EL(Y, g(\mathbf{X})).$$

If a squared error loss $L(Y, g(\mathbf{X})) = (y - g(\mathbf{X}))^2$ is given, then

$$f(\mathbf{x}) = E(Y|\mathbf{X} = \mathbf{x})$$

Let $\hat{f}$ be the estimate of f based on the training data $\chi_n = \{(\mathbf{X}_i, Y_i), i = 1, \dots, n\}$. Then, the prediction error based on the training data $T = (\mathbf{Z}, W)$ is given by

$$PE = E_{\chi_n} E_{\mathbf{Z}, W} L(W, \hat{f}(\mathbf{Z})).$$

Now, we apply the prediction error to the linear regression model.

Assume that the model is

$$Y = \mu + \varepsilon,$$

where $E(\varepsilon) = 0$, $Var(\varepsilon) = \sigma^2$. Let $\hat{f}(\mathbf{x}) = \mathbf{x}'\hat{\boldsymbol{\beta}}$, where $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is based on the training data $\chi_n = \{(\mathbf{X}_i, Y_i), i = 1, \dots, n\}$. Now, for the test data $T = (\mathbf{Z}, W)$, the prediction error can be written as

$$\begin{aligned} PE &= E_{\mathbf{Z}, W} E_{\chi_n} (W - \mathbf{z}'\hat{\boldsymbol{\beta}})^2 \\ &= E_{\mathbf{Z}, W} E_{\chi_n} (W - \mu + \mu - \mathbf{z}'\hat{\boldsymbol{\beta}})^2 \\ &= E_{\mathbf{Z}, W} E_{\chi_n} \{(W - \mu)^2 + (\mu - \mathbf{z}'\hat{\boldsymbol{\beta}})^2 + 2(W - \mu)(\mu - \mathbf{z}'\hat{\boldsymbol{\beta}})\} \\ &= \sigma^2 + E_{\mathbf{Z}, W} E_{\chi_n} (\mu - \mathbf{z}'\hat{\boldsymbol{\beta}})^2, \end{aligned}$$

where σ^2 is called *irreducible error*, and the 2nd term is called *ME(model error)*.

Therefore, minimizing the prediction error is equivalent to minimizing the model error. Now, we let (X, w) be the test data, then

$$\begin{aligned}
ME &= E[(\mu - X'\hat{\beta})'(\mu - X'\hat{\beta})] \\
&= E[(\mu - Hy)'(\mu - Hy)] \\
&= E[(\mu - H\mu - H\varepsilon)'(\mu - H\mu - H\varepsilon)] \\
&= E[\{(I - H)\mu - H\varepsilon\}'\{(I - H)\mu - H\varepsilon\}] \\
&= E[\mu'(I - H)\mu + \varepsilon'H\varepsilon] \\
&= \mu'(I - H)\mu + p\sigma^2,
\end{aligned}$$

where $H = X(X'X)^{-1}X'$. Note that $Bias(X'\hat{\beta}) = E(X'\hat{\beta}) - \mu$, so that

$$\mu'(I - H)\mu = Bias(X'\hat{\beta})'Bias(X'\hat{\beta})$$

and

$$p\sigma^2 = \text{tr}[Cov(X'\hat{\beta})].$$

Therefore, the model error is the same as the MSE (mean squared error).

5.1.3 Example for Model Selection

Let $\mu \equiv \mu(x) = \exp(0.1 + 0.02x^2)$ be a true model, and we generate 100 random numbers $(x_i, y_i), i = 1, \dots, 100$, where $x = 1(0.09)10$ and

$\varepsilon_i \sim N(0, 1.5^2)$, i.e.,

$$Y_i = \exp(0.1 + 0.02x_i^2) + \varepsilon_i$$

and our postulated model is

$$f_6(x) = \sum_{j=0}^6 \beta_j x^j$$

See Fig. 5.1 (p. 207); $\mu(5) = \exp(0.1 + 0.02 \times 5^2) = 1.822$ vs $\hat{f}_j(5)$

5.2 Criteria for Variable Selection

Assume that the number of all possible candidate covariates is $k - 1$.

- Types of models

(i) *maximal (full, saturated)* model : use $k - 1$ variables in fit

(ii) *current* model : use $p - 1 (p < k)$ variables

(iii) *minimal (null)* model : do not use any covariates

Ex. 5.1 (p.208) : house price data (X_1, X_2, X_3, X_4)

-Fit using the full model

$$\hat{Y} = 1.22 + .0519X_1 + .0116X_2 + .349X_3 - .219X_4$$

with $R^2 = 0.93$.

5.2.1 R_p^2

$$R_p^2 = \frac{SSR_p}{SST} = 1 - \frac{SSE_p}{SST}$$

See Fig. 5.3 (p. 211)

5.2.2 R_{ap}^2 and s_p^2

$$R_{ap}^2 = 1 - \frac{SSE_p / (n - p)}{SST / (n - 1)} = 1 - \frac{s_p^2}{SST / (n - 1)}$$

See Fig. 5.4 (p. 212)

5.2.3 Mallows' C_p

(1) Motivation

Minimize total mean squared errors

(2) Derivation of C_p

Assume that the true regression model is

$$\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon},$$

where $E(\mathbf{y}) = \boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$. On the other hand, the postulated model is $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Let $\hat{\mathbf{y}}$ be the fitted vector under the postulated model, then the mean squared error of the i th observation is

$$E(\hat{y}_i - \mu_i)^2$$

which can be rewritten as

$$\begin{aligned} E(\hat{y}_i - \mu_i)^2 &= E\{(\hat{y}_i - E(\hat{y}_i)) + (E(\hat{y}_i) - \mu_i)\}^2 \\ &= \text{Var}(\hat{y}_i) + \{E(\hat{y}_i) - \mu_i\}^2, \end{aligned}$$

i.e., $MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2(\hat{\theta})$. Now, we want to minimize the total MSE,

$$\begin{aligned} \Gamma_p &= \sum_{i=1}^n \{\text{Var}(\hat{y}_i) + (E(\hat{y}_i) - \mu_i)^2\} / \sigma^2 \\ &= \frac{1}{\sigma^2} \{\text{tr}[\text{Cov}(\hat{\mathbf{y}})] + (E(\hat{\mathbf{y}}) - \boldsymbol{\mu})'(E(\hat{\mathbf{y}}) - \boldsymbol{\mu})\}. \end{aligned}$$

Note that

$$\text{tr}[\text{Cov}(\hat{\mathbf{y}})] = \text{tr}[\text{Cov}(\mathbf{H}\mathbf{y})] = \text{tr}[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \sigma^2] = p\sigma^2$$

and

$$E(\hat{\mathbf{y}}) - \boldsymbol{\mu} = E(\mathbf{H}\mathbf{y}) - \boldsymbol{\mu} = -(\mathbf{I} - \mathbf{H})\boldsymbol{\mu}$$

Hence, we have

$$\Gamma_p = p + \frac{1}{\sigma^2} \boldsymbol{\mu}'(\mathbf{I} - \mathbf{H})\boldsymbol{\mu}$$

which contains unknown parameters such as σ^2 , $\boldsymbol{\mu}$, we need to estimate them. Note that

$$E(SSE_p) = E(\mathbf{e}'\mathbf{e}) = E[\mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y}] = \boldsymbol{\mu}'(\mathbf{I} - \mathbf{H})\boldsymbol{\mu} + (n - p)\sigma^2,$$

where SSE_p is the residual sum of squares under the current model with $p - 1$ covariates. Therefore,

$$\frac{E(SSE_p)}{\sigma^2} - (n - 2p) = \Gamma_p$$

Hence, as an estimator for Γ_p , Mallows (1973) suggested to use

$$C_p = \frac{SSE_p}{s^2} - (n - 2p),$$

where s^2 is an estimator of σ^2 under the maximal model.

(3) Properties of C_p

(i) C_p = Goodness-of-fit + Penalty term

1st term = SSE_p/s^2 : decreases as the number of covariates increases

2nd term = $2p - n$: increases as the number of covariates increases

Hence, C_p is often called *penalized criterion*.

(ii) $C_p \simeq p$

If the current model is good, then $SSE_p \simeq (n - p)s^2$, and therefore, $C_p \simeq (n - p) - (n - 2p) = p$. In fact, under the maximal model, we have $SSE_p = (n - p)s^2$, so that $C_p = p$.

See Fig. 5.5 (p. 215)

5.2.4 $PRESS_p$

(1) Motivation

Minimize the PRESS (PREdiction Sum of Squared errors) suggested by Allen (1971). Use $n - 1$ observations as training set, and use 1 observation as prediction (test set).

(2) Derivation

$$PRESS_p = \sum_{i=1}^n (y_i - \hat{y}_{i(i)})^2,$$

where $\hat{y}_{i(i)}$ is the fitted value at $\mathbf{x}_i$ based on $n - 1$ observations. Note that

$$\hat{y}_{i(i)} = \mathbf{x}_i' \hat{\boldsymbol{\beta}}_{(i)} = \mathbf{x}_i' \left(\hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - h_{ii}} \right) = \hat{y}_i - \frac{h_{ii} e_i}{1 - h_{ii}}$$

and therefore,

$$PRESS_p = \sum_{i=1}^n \left(y_i - \hat{y}_i + \frac{h_{ii} e_i}{1 - h_{ii}} \right)^2 = \sum_{i=1}^n \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

See Fig. 5.6 (p. 216)

5.3 Methods of Variable Selection

- (i) (all possible regression)
- (ii) (forward stepwise regression)
- (iii) (forward selection)
- (iv) (backward elimination)

5.3.1 All Possible Regression

Assume that the maximal model has $k - 1$ covariates. Then the number of possible models are

$$\binom{k-1}{0} + \binom{k-1}{1} + \cdots + \binom{k-1}{k-1} = 2^{k-1}$$

which is almost computationally infeasible if k is large.

5.3.2 Forward stepwise regression

Algorithm

STEP 1. Assume that the current model has 1 covariate, and make an F -test

$$F_i^* = \frac{MSR_i}{s_i^2}, \quad i = 1, \dots, k-1$$

and if $\max_{1 \leq i \leq k-1} F_i^* > F_\alpha(1, n-2)$, then the corresponding variable is entered. Otherwise, select the null model.

STEP 2. Assume that X_7 has been selected in STEP 1. Do the partial F -test

for other variables than X_7 , i.e.,

$$F_i^* = \frac{MSR(X_i | X_7)}{s^2(X_7, X_i)} = \left(\frac{\hat{\beta}_i}{SE(\hat{\beta}_i)} \right)^2, \quad i \neq 7$$

and if $\max_{i \neq 7} F_i^* > F_\alpha(1, n-3)$, then the corresponding variable is entered.

Otherwise, stop.

STEP 3. Assume that X_3 has been selected in STEP 2. Do the partial F -test for X_7 when X_3 is given, i.e.,

$$F_7^* = \frac{MSR(X_7 | X_3)}{s^2(X_3, X_7)}$$

and if is larger than $F_\alpha(1, n - 3)$, then X_7 remains in the model. Otherwise, X_7 is deleted from the model.

STEP 4. If X_7 remain the model in STEP 3, then repeat STEP 2 and STEP 3.

Ex. 5.2 (p.219)

5.3.3 Forward selection

Start with the null model, and the variable with largest F -value is entered if it is significant. We continue this process until no further significant variable appears. Note that if some variable is entered, it never removed from the model.

5.3.4 Backward elimination

Start with the maximal model, and the variable with smallest F -value is removed if it is not significant. We continue this process until no further non-significant variable appears. Note that if some variable is removed, it never remains in the model.

$$F_1^* = \frac{MSR(X_1 | X_2, \dots, X_{k-1})}{s^2(X_1, \dots, X_{k-1})}$$

5.4 Model Validation

5.4.1 Cross validation

(1) Motivation

For the estimation of prediction error, the *cross – validation*(CV) is often used. The idea is partition n observations into two sets; one is the training set and the other is the test set.

(2) Definition

Partition a set $\{1, \dots, n\}$ into K sets $I_j, j = 1, \dots, K$, and define a function $\kappa : \{1, \dots, n\} \mapsto \{1, \dots, K\}$ such that $\kappa(i) = j \Leftrightarrow i \in I_j$. For the data set $\{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$, let $\hat{f}_{-j}$ be the estimator based on observations except in I_j . Then, the cross-validation estimation for the prediction error is defined as

$$\hat{P}E_{CV} = \frac{1}{n} \sum_{i=1}^n L(Y_i, \hat{f}_{-\kappa(i)}(\mathbf{X}_i)),$$

where L is a loss function.

(3) k -fold CV

When we partition n observations into k groups with equal size, i.e., $n = kc$ for some c , then k -fold cross validation uses $k - 1$ group for the training set and one group for the test set. As a special case, if $K = n$ and $c = 1$, then it is called LOOCV (leave-one-out-cross-validation), i.e., 1-fold CV. It is recommended that either $k = 5$ or $k = 10$.

(4) Example

Assume that $n = 6$, $k = 3$, and 3 groups are $\{1, 5\}$, $\{2, 6\}$, $\{3, 4\}$.

training data	test data
1, 2, 5, 6	3, 4
1, 3, 4, 5	2, 6
2, 3, 4, 6	1, 5

5.4.2 Training data, test data, and prediction error

See Fig. 5.8 (p. 222)

5.5 Diagnostics on the Variable Selection

Let $K = \{i_1, \dots, i_k\}$ be an index set of size k . Here, we want to see the effect of k observations in K on Mallows' C_p . Recall that

$$C_p = \frac{SSE_p}{s^2} - (n - 2p).$$

Let $\mathbf{e}^c$ and $\mathbf{e}$ be the residual vectors under the current model and the maximal model, respectively. If $q - 1$ is the number of covariates under the

maximal model, then

$$C_p = \frac{\mathbf{e}^{c'} \mathbf{e}^c}{\mathbf{e}' \mathbf{e} / (n - q)} - (n - 2p).$$

Now, the C_p based on $n - k$ observations after deleting k observations in K is

$$C_{p(K)} = \frac{SSE_{p(K)}}{s_{(K)}^2} - (n - k - 2p)$$

and it can be shown that

$$C_{p(K)} = \frac{(n - k - q) \{C_p + (n - 2p) - \mathbf{e}_K^{c'} (\mathbf{I} - \mathbf{H}_K^c)^{-1} \mathbf{e}_K^c / s^2\}}{(n - q) - \mathbf{e}_K (\mathbf{I} - \mathbf{H}_K)^{-1} \mathbf{e}_K / s^2} - (n - k - 2p)$$

Ex. 5.3 (p.224)

5.6 Miscellanea in Model Selection

5.6.1 Jensen inequality and the Kullback-Leibler distance

Theorem 5.1 (Jensen's inequality) Let ϕ be a convex function in an open interval I , and assume that $P(X \in I) = 1$ and $E(X) < \infty$. Then, we have

$$\phi[E(X)] \leq E[\phi(X)]$$

which is called the Jensen's inequality.

Definition 5.1 (Kullback-Leibler distance) Assume that two pdfs f and g are given. Then, the Kullback-Leibler distance between g and f , Kullback-Leibler information of g at f , or the entropy distance between f and g with

respect to f is

$$E_f \left[\log \left(\frac{f(X)}{g(X)} \right) \right] = \int \log \left(\frac{f(x)}{g(x)} \right) f(x) dx$$

As a corollary, we have

$$\begin{aligned} E_f \left[\log \left(\frac{f(X)}{g(X)} \right) \right] &= -E_f \left[\log \left(\frac{g(X)}{f(X)} \right) \right] \\ &\geq -\log \left[E_f \left(\frac{g(X)}{f(X)} \right) \right] \\ &= 0. \end{aligned}$$

Note that Kullback-Leibler distance is always non-negative, and 0 if $f = g$.

5.6.2 Akaike information criterion

Let $\theta_0 \in R^p$ be the true model and $\theta \in R^p$ be the postulated model.

Also, let $f(x) \equiv f(x : \theta_0)$ and $f(x : \theta)$ be the pdfs under the true and postulated model, respectively. Now, let

$$K(\theta) \equiv -2E_{\theta_0}[\log f(X : \theta)] = -2 \int \log f(x : \theta) f(x) dx$$

and consider

$$\frac{1}{2}K(\theta) + \int \{\log f(x)\} f(x) dx$$

which is Kullback-Leibler distance because if we let $g(x) \equiv f(x : \theta)$, then

$$\begin{aligned} \frac{1}{2}K(\theta) + \int \{\log f(x)\}f(x)dx &= - \int \log f(x : \theta)f(x)dx + \int \{\log f(x)\}f(x)dx \\ &= \int \log \left(\frac{f(x)}{g(x)} \right) f(x)dx \\ &= E_f \left[\log \left(\frac{f(x)}{g(x)} \right) \right]. \end{aligned}$$

The AIC (Akaike Information Criterion, Akaike 1973) minimizes $K(\theta)$, however, it contains unknown parameter $f(x) \equiv f(x : \theta_0)$, so that we need to estimate it. As an estimator for $K(\theta)$, we consider

$$K_n(\theta) = -\frac{2}{n} \sum_{i=1}^n \log f(X_i : \theta).$$

Let $\dot{K}_n(\theta) = \partial K_n(\theta) / \partial \theta$, $\ddot{K}_n(\theta) = \partial^2 K_n(\theta) / \partial \theta \partial \theta'$, and let $\hat{\theta}$ be the MLE of θ , then $\dot{K}_n(\hat{\theta}) = 0$, and let

$$K_n(\hat{\theta}) = -\frac{2}{n} \sum_{i=1}^n \log f(X_i : \hat{\theta}).$$

Finally, the AIC is defined as the sum of $K_n(\hat{\theta})$ and $\frac{2p}{n}$, i.e.,

$$AIC = -\frac{2}{n} \sum_{i=1}^n \log f(X_i : \hat{\theta}) + \frac{2p}{n}$$

which is often used as a model selection criterion.

5.6.3 Justification of AIC

Here, we show that AIC is a good estimator for $K(\theta)$ by the following;

Theorem 5.2

$$K_n(\hat{\boldsymbol{\theta}}) = K(\hat{\boldsymbol{\theta}}) + R_n + o_p(n^{-1}),$$

where $E(R_n) = -2p/n + o(n^{-1})$.

(Proof) By noting that $\dot{K}_n(\hat{\boldsymbol{\theta}}) = 0$, and by the Taylor expansion, we have

$$\begin{aligned} K_n(\boldsymbol{\theta}_0) &= K_n(\hat{\boldsymbol{\theta}}) + (\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}})' \dot{K}_n(\hat{\boldsymbol{\theta}}) + \frac{1}{2}(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}})' \ddot{K}_n(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}) + o_p(n^{-1}) \\ &= K_n(\hat{\boldsymbol{\theta}}) + \frac{1}{2}(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}})' \ddot{K}_n(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}) + o_p(n^{-1}). \end{aligned}$$

Also, by the Taylor expansion of $\ddot{K}_n(\hat{\boldsymbol{\theta}})$,

$$\ddot{K}_n(\hat{\boldsymbol{\theta}}) = \ddot{K}_n(\boldsymbol{\theta}_0) + o_p(1) = 2I(\boldsymbol{\theta}_0) + o_p(1),$$

where

$$I(\boldsymbol{\theta}_0) = -E_{\boldsymbol{\theta}_0}[\partial^2 \log f(X_1 : \boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}']$$

is the Fisher information. Therefore,

$$K_n(\hat{\boldsymbol{\theta}}) = K_n(\boldsymbol{\theta}_0) - (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' I_n(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(n^{-1})$$

and further,

$$\begin{aligned} K(\hat{\boldsymbol{\theta}}) &= K(\boldsymbol{\theta}_0) + (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' \int \left[-2 \frac{\partial}{\partial \boldsymbol{\theta}} \log f(x : \boldsymbol{\theta}) \right]_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} f(x : \boldsymbol{\theta}_0) dx \\ &\quad + (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' \left\{ \int \left[-\frac{\partial^2 \log f(x : \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right]_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} f(x : \boldsymbol{\theta}_0) dx \right\} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(n^{-1}) \end{aligned}$$

Hence,

$$\begin{aligned} K_n(\hat{\boldsymbol{\theta}}) &= K(\hat{\boldsymbol{\theta}}) + \{K_n(\boldsymbol{\theta}_0) - K(\boldsymbol{\theta}_0) - 2(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' I_n(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\} + o_p(n^{-1}) \\ &\equiv K(\hat{\boldsymbol{\theta}}) + R_n + o_p(n^{-1}) \end{aligned}$$

where $R_n = K_n(\boldsymbol{\theta}_0) - K(\boldsymbol{\theta}_0) - 2(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' I_n(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$. Now, by the asymptotic normality of the MLE,

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow N_p(\mathbf{0}, I^{-1}(\boldsymbol{\theta}_0))$$

we have

$$n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' I(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow \chi_p^2$$

and

$$E_{\boldsymbol{\theta}_0} [n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' I(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)] \rightarrow p$$

Finally, by the WLLN (weak law of large numbers), we have

$$K_n(\boldsymbol{\theta}_0) \xrightarrow{p} K(\boldsymbol{\theta}_0).$$

5.6.4 Application of AIC to model selection

Consider a multiple linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}),$$

where σ^2 is unknown. Now,

$$\begin{aligned} K_n(\hat{\theta}) &= -\frac{2}{n} \sum \log f(X_i : \hat{\theta}) \\ &= -\frac{2}{n} \left\{ -\frac{1}{2\sigma^2} \sum (Y_i - x_i' \hat{\beta})^2 \right\} + \text{const} \\ &= SSE_p / n\sigma^2 + \text{const} \end{aligned}$$

so that

$$AIC = \frac{SSE_p}{n\sigma^2} + \frac{2p}{n} = \frac{1}{n} C_p + \text{const}$$

Therefore, AIC gives the same result as the C_p . Also, BIC (Bayesian Information Criterion) is defined as

$$BIC = \frac{SSE_p}{n\sigma^2} + \frac{p \log n}{n}.$$

Note that BIC gives more penalty to model complexity than AIC, it tends to select simpler model than AIC.

Chapter 6. Transformation of the Linear Regression Model

6.1 The Use of Dummy Variables

6.1.1 One dummy variable case

(1) Motivating example

Problem : A company make advertisements either in the newspaper or the broadcast. They want to see the difference in the advertisement effect between the newspaper and the broadcast.

Solution : Let

$$Z = \begin{cases} 1 & \text{if newspaper} \\ 0 & \text{if broadcast} \end{cases}$$

and consider

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon,$$

where Z is called *dummy variable*(*indicator variable*). Hence,

$$Z = 0 : E[Y] = \beta_0 + \beta_1 X$$

$$Z = 1 : E[Y] = (\beta_0 + \beta_2) + \beta_1 X$$

Test : $H_0 : \beta_2 = 0$

Remark. If a categorical variable has c categories, then we must have $c - 1$ dummy variables. For example, in this problem, we have two categories, so that we need $1 (= 2 - 1)$ dummy variable. Now, assume that we defined two dummy variables in this example, i.e.,

$$Z_1 = \begin{cases} 1 & \text{if newspaper} \\ 0 & \text{otherwise} \end{cases}$$

$$Z_2 = \begin{cases} 1 & \text{if broadcast} \\ 0 & \text{otherwise} \end{cases}$$

Then, the corresponding model would be

$$Y = \beta_0 + \beta_1 X + \beta_2 Z_1 + \beta_3 Z_2 + \varepsilon$$

then the design matrix might be, for example,

$$\mathbf{X} = \begin{bmatrix} 1 & X_1 & 1 & 0 \\ 1 & X_2 & 1 & 0 \\ 1 & X_3 & 1 & 0 \\ 1 & X_4 & 1 & 0 \\ 1 & X_5 & 1 & 0 \\ 1 & X_6 & 0 & 1 \\ 1 & X_7 & 0 & 1 \\ 1 & X_8 & 0 & 1 \\ 1 & X_9 & 0 & 1 \\ 1 & X_{10} & 0 & 1 \end{bmatrix}$$

which is not a full rank matrix because the sum of the 3rd and 4th columns

become the 1st column. Therefore, $\mathbf{X}'\mathbf{X}$ is not invertible.

(2) Example

We have 4 types of seed; A, B, C, and D, and want to see the difference in products between seeds. Since this categorical variable has 4 categories, we must define $3(= 4 - 1)$ dummy variables. Hence, define

$$Z_1 = \begin{cases} 1 & \text{if A} \\ 0 & \text{otherwise} \end{cases}$$

$$Z_2 = \begin{cases} 1 & \text{if B} \\ 0 & \text{otherwise} \end{cases}$$

$$Z_3 = \begin{cases} 1 & \text{C} \\ 0 & \text{otherwise} \end{cases}$$

Note that seed D is given when $Z_1 = Z_2 = Z_3 = 0$. Hence, the model becomes

$$Y = \beta_0 + \beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3 + \varepsilon$$

and the expectation of product of each seed is

$$A : E[Y] = \beta_0 + \beta_1$$

$$B : E[Y] = \beta_0 + \beta_2$$

$$C : E[Y] = \beta_0 + \beta_3$$

$$D : E[Y] = \beta_0.$$

For example, the difference of products between A and B is $\beta_2 - \beta_1$.

6.1.2 Extension of the use of dummy variables

(1) Interaction term

If we include the interaction term, then

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \varepsilon$$

so that

$$Z = 0 : E[Y] = \beta_0 + \beta_1 X$$

$$Z = 1 : E[Y] = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)X$$

Here, both the slope and the intercept are different.

(2) Break (change) point

If we have a prior information on the change point, for example,

$$Z = \begin{cases} 1 & X > 20 \\ 0 & X \leq 20 \end{cases}$$

then, we can model like

$$Y = \beta_0 + \beta_1 X + \beta_2 (X - 20)Z + \varepsilon$$

so that

$$1. X \leq 20 : E[Y] = \beta_0 + \beta_1 X$$

$$2. X > 20 : E[Y] = (\beta_0 - 20\beta_2) + (\beta_1 + \beta_2)X$$

6.1.3 ANOVA using the regression model

Consider the following oneway ANOVA model

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2, \dots, c; \quad j = 1, 2, \dots, n_i,$$

where Y_{ij} is the j th response of the i th group, μ is a overall effect, α_i is the effect of the i th group, and ε_{ij} is error term. For $c = 4$ and $n_i = 3$, we have

$$\boldsymbol{\beta} = (\mu, \alpha_1, \alpha_2, \alpha_3, \alpha_4)'$$

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

But, $\mathbf{X}$ is not a full rank matrix, so that it is not estimable. To overcome this difficulty, use dummy variables. Let

$$Z_1 = \begin{cases} 1 & \text{group 1} \\ 0 & \text{otherwise} \end{cases}$$

$$Z_2 = \begin{cases} 1 & \text{group 2} \\ 0 & \text{otherwise} \end{cases}$$

$$Z_3 = \begin{cases} 1 & \text{group 3} \\ 0 & \text{otherwise} \end{cases}$$

Then,

$$Y_i = \beta_0 + \beta_1 Z_{i1} + \beta_2 Z_{i2} + \beta_3 Z_{i3} + \varepsilon_i, \quad i = 1, 2, \dots, 12$$

Note that the test for $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$ is equivalent to

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0.$$

Ex. 6.3 (p.242)

6.2 Polynomial Regression

1st order polynomial regression

$$1 \text{ covariate : } Y = \beta_0 + \beta_1 X + \varepsilon$$

$$2 \text{ covariates : } Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

$$k \text{ covariates : } Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \varepsilon$$

2nd order polynomial regression

$$1 \text{ covariate : } Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

$$2 \text{ covariates : } Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 + \varepsilon$$

$$k \text{ covariates : } Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum_{i < j} \beta_{ij} X_i X_j + \varepsilon$$

6.2.1 k th order polynomial regression

The k th order polynomial regression with 1 covariate is

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_k X^k + \varepsilon$$

(1) Estimation : Since the column vectors of X in the polynomial regression are highly correlated, we need to transform covariates. Among them orthogonal polynomial is often used, i.e.,

$$Y_i = \alpha_0 \phi_0(X_i) + \alpha_1 \phi_1(X_i) + \cdots + \alpha_k \phi_k(X_i) + \varepsilon_i,$$

where $\phi_r(X)$ is a r th degree orthogonal polynomial such that $\sum_{i=1}^n \phi_r(X_i) \phi_s(X_i) = 0$, $\forall r \neq s$. Then,

$$\hat{\alpha}_r = \frac{\sum_{i=1}^n \phi_r(X_i) Y_i}{\sum_{i=1}^n \phi_r^2(X_i)}, \quad r = 0, 1, \dots, k$$

(2) Determination of k : Sequential test for the degree is often used, i.e.,

STEP 1 : Fit $Y = \beta_0 + \beta_1 X + \varepsilon$, and test $H_0 : \beta_1 = 0$. If not rejected, then stop, i.e., $k = 1$. If rejected, go to STEP 2.

STEP 2 : Fit $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$, and test $H_0 : \beta_2 = 0$. If not rejected, then stop, i.e., $k = 2$. If rejected, go to STEP 3.

Ex. 6.4 (p.246)

6.2.2 Response surface analysis

Goal : analysis of response surface and finding the optimal condition for the response

(1) 2nd order model with 1 covariate case

Consider a fitted 2nd order model

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$$

Then, the optimal condition can be obtained by the 1st derivative, i.e.,

$$\frac{d\hat{Y}}{dX} = \hat{\beta}_1 + 2\hat{\beta}_2 X = 0$$

and we obtain

$$X_m = -\frac{\hat{\beta}_1}{2\hat{\beta}_2}$$

which is called a stationary point and $\hat{Y}$ at this point becomes

$$\hat{Y}_m = \hat{\beta}_0 - \frac{\hat{\beta}_1^2}{4\hat{\beta}_2}$$

Also, the 2nd derivative is

$$\frac{d^2\hat{Y}}{dX^2} = 2\hat{\beta}_2.$$

Therefore, if $\hat{\beta}_2 > 0$, then $\hat{Y}_m$ is minimum, and if $\hat{\beta}_2 < 0$, then $\hat{Y}_m$ is maximum.

(2) 1st order model with 2 covariates case

Suppose that we have

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2.$$

Here, we cannot obtain stationary points X_1 and X_2 where Y is optimal, and we can only obtain the increasing (decreasing) direction. Hence, for a fixed Y , we have

$$X_2 = -\frac{\hat{\beta}_1}{\hat{\beta}_2} X_1 + c$$

Therefore, $\hat{\beta}_2 / \hat{\beta}_1$, the perpendicular direction of the slope, gives the maximum increase (decrease), and it is called *steepest ascent (descent)*.

(3) 2nd order model with k covariates case

Suppose that we have

$$Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \sum_{i=1}^k \beta_{ii} X_i^2 + \sum_{i < j}^k \beta_{ij} X_i X_j + \varepsilon$$

In matrix notation,

$$Y = \beta_0 + \mathbf{x}'\boldsymbol{\beta} + \mathbf{x}'\mathbf{B}\mathbf{x} + \varepsilon,$$

where $\mathbf{x} = (X_1, X_2, \dots, X_k)'$, $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_k)'$, and

$$\mathbf{B} = \begin{bmatrix} \beta_{11} & \frac{\beta_{12}}{2} & \cdots & \frac{\beta_{1k}}{2} \\ \frac{\beta_{12}}{2} & \beta_{22} & \cdots & \frac{\beta_{2k}}{2} \\ \vdots & \vdots & & \vdots \\ \frac{\beta_{1k}}{2} & \frac{\beta_{2k}}{2} & \cdots & \beta_{kk} \end{bmatrix}$$

Let the fitted model is

$$\hat{Y} = \hat{\beta}_0 + \mathbf{x}'\hat{\boldsymbol{\beta}} + \mathbf{x}'\hat{\mathbf{B}}\mathbf{x}$$

and the 1st derivative gives

$$\frac{\partial \hat{Y}}{\partial \mathbf{x}} = \hat{\boldsymbol{\beta}} + 2\hat{\mathbf{B}}\mathbf{x} = \mathbf{0}$$

so that the stationary point is

$$\mathbf{x}_s = -\frac{1}{2}\hat{\mathbf{B}}^{-1}\hat{\boldsymbol{\beta}}$$

The analysis of response at the stationary point is called *canonical analysis*, and two methods, eigenvalue analysis and contour plot, are often used for the canonical analysis. Now, the fitted value at the stationary point is

$$\hat{Y}_s = \hat{\beta}_0 + \mathbf{x}_s'\hat{\boldsymbol{\beta}} + \mathbf{x}_s'\hat{\mathbf{B}}\mathbf{x}_s$$

and let $w = x - x_s$. Then, we have

$$\begin{aligned}
 \hat{Y} &= \hat{\beta}_0 + (x_s + w)' \hat{\beta} + (x_s + w)' \hat{B}(x_s + w) \\
 &= \hat{\beta}_0 + x_s' \hat{\beta} + x_s' \hat{B} x_s + w' \hat{\beta} + 2w' \hat{B} x_s + w' \hat{B} w \\
 &= \hat{Y}_s + w' \hat{B} w + w' (\hat{\beta} + 2\hat{B} x_s) \\
 &= \hat{Y}_s + w' \hat{B} w
 \end{aligned}$$

Therefore, if $\hat{B}$ is positive definite, then $\hat{Y}_s$ is minimum, and if $\hat{B}$ is negative definite, then $\hat{Y}_s$ is maximum. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be eigenvalues of $\hat{B}$. Then, we have

1. If all the λ'_j s are positive, then $\hat{Y}_s$ is minimum.
2. If all the λ'_j s are negative, then $\hat{Y}_s$ is maximum.
3. If some λ'_j s are positive and some λ'_j s are negative, then $\hat{Y}_s$ is neither minimum or maximum. The corresponding x_s is called a *saddle point*.

Fig. 6.5 (p. 250)

Ex. 6.5 (p.251)

6.3 Weighted Least Squares Method

So far we assumed $Cov(\varepsilon) = I\sigma^2$ which is called *homogeneity* assumption.

If the homogeneity assumption is false, then we often assume that

$$Cov(\varepsilon) = \sigma^2 \mathbf{W}^{-1},$$

where $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$, and it is called *heteroscedasticity*. Therefore, $Var(\varepsilon_i) = \sigma^2 / w_{ii}$. Now, there exists a non-singular matrix $\mathbf{P}$ s.t.

$$\mathbf{P}^2 = \mathbf{W}.$$

Let $\delta = \mathbf{P}\varepsilon$, then $E(\delta) = 0$ and

$$Cov(\delta) = E(\delta\delta') = E(\mathbf{P}\varepsilon\varepsilon'\mathbf{P}) = \mathbf{P}\mathbf{W}\mathbf{P}\sigma^2 = I\sigma^2$$

Now, in $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, multiply $\mathbf{P}$ on both sides, then

$$\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{X}\beta + \mathbf{P}\varepsilon$$

and let $\mathbf{P}\mathbf{y} = \mathbf{z}$, $\mathbf{P}\mathbf{X} = \mathbf{Q}$, then

$$\mathbf{z} = \mathbf{Q}\beta + \delta$$

Hence, the normal equation becomes

$$\mathbf{Q}'\mathbf{Q}\hat{\beta} = \mathbf{Q}'\mathbf{y}$$

and

$$X'WX\hat{\beta} = X'Wy$$

gives

$$\hat{\beta} = (X'WX)^{-1}X'Wy.$$

The above method is called *weighted* least squares method.

Ex. 6.6 (p.254)

6.4 Box-Cox Transformation Model

So far we assumed the normality of response, i.e. $\varepsilon \sim N(\mathbf{0}, \sigma^2 I)$. Hence, when the normality assumption is doubtful, Box and Cox (1964) suggested the followings; Box and Cox argued that there exists λ s.t.

$$w_i = \begin{cases} (y_i^\lambda - 1) / \lambda, & \lambda \neq 0 \\ \log y_i, & \lambda = 0 \end{cases}$$

is approximately normal. In matrix notation,

$$w(\lambda) = X\beta + \varepsilon$$

which is called *Box – Cox transformation model*. Here, we need to estimate λ in addition to β and σ^2 , and we will obtain the MLEs for them. Now, the

likelihood function is

$$L = (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} (\mathbf{w} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{w} - \mathbf{X}\boldsymbol{\beta}) \right] J(\lambda)$$

where

$$J(\lambda) = \prod_{i=1}^n \frac{\partial w_i}{\partial y_i} = \prod_{i=1}^n y_i^{\lambda-1}$$

is the Jacobian from transforming y to $w(\lambda)$. To get the MLE of λ , $\boldsymbol{\beta}$, and σ^2 , we first fix λ , and get the MLE for $\boldsymbol{\beta}$ and σ^2 . Then,

$$\hat{\boldsymbol{\beta}}(\lambda) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{w}(\lambda), \quad s^2(\lambda) = \mathbf{w}(\lambda)'(\mathbf{I} - \mathbf{H})\mathbf{w}(\lambda)/n$$

Now, we replace $\hat{\boldsymbol{\beta}}(\lambda)$ and $s^2(\lambda)$ in the likelihood, then the likelihood becomes a function of λ only, i.e.,

$$\begin{aligned} l(\lambda; \hat{\boldsymbol{\beta}}(\lambda), s^2(\lambda)) &= -\frac{n}{2} \log s^2(\lambda) + \log J(\lambda) \\ &= -\frac{n}{2} \log s^2(\lambda) + (\lambda - 1) \sum \log y_i \end{aligned}$$

Now, to get the MLE of λ , we need to solve $\partial l(\lambda; \hat{\boldsymbol{\beta}}(\lambda))/\partial \lambda = 0$, however, it does not give an explicit solution because it is nonlinear in λ . In this case, we are only to obtain the numerical result for the MLE through the grid search, say.

Ex. 6.7 (p.259)

6.5 Robust Regression

The distribution of error terms might have heavier tails than the normal distribution. For example, the double exponential (Laplace) distribution whose pdf is given by

$$f(\varepsilon) = \frac{1}{2\sigma} e^{-|\varepsilon|/\sigma}, \quad -\infty < \varepsilon < \infty$$

Also, the likelihood function becomes

$$\frac{1}{(2\sigma)^n} \exp[-\sum |y_i - \mathbf{x}'_i \boldsymbol{\beta}| / \sigma]$$

Now, to estimate $\boldsymbol{\beta}$, maximizing the likelihood is equivalent to minimizing $\sum |y_i - \mathbf{x}'_i \boldsymbol{\beta}|$, which is called L_1 -norm regression. In the normal distribution, we minimize $\sum (y_i - \mathbf{x}'_i \boldsymbol{\beta})^2$, which is L_2 -norm regression. In general, L_2 -norm regression minimizes $\sum (y_i - \mathbf{x}'_i \boldsymbol{\beta})^p$, $1 \leq p \leq 2$.

Fig. 6.10 (p. 261)

6.5.1 M-Estimation

For some function $\rho(\cdot)$, if $\boldsymbol{\beta}$ is obtained by

$$\min \sum_{i=1}^n \rho(\varepsilon_i) = \min \sum_{i=1}^n \rho(y_i - \mathbf{x}'_i \boldsymbol{\beta})$$

then it is called M -estimator. If $\rho(u) = \frac{1}{2}u^2$, then it is L_2 -norm regression.

Also, if $\rho(u) = |u|$, then it is L_1 -norm regression.

Now, to obtain the robust estimator of β , we consider

$$\min \sum_{i=1}^n \rho \left(\frac{\varepsilon_i}{s} \right) = \min \sum_{i=1}^n \rho \left(\frac{y_i - \mathbf{x}_i' \beta}{s} \right)$$

where s is a robust estimator of σ , and it is given by

$$s = \text{median}|e_i - \text{median}(e_i)| / 0.6745$$

We can show that s is an unbiased estimator of σ if n is large and error terms have normal distribution. To obtain the minimum, we take 1st derivative of ρ w.r.t. β_j , $j = 0, 1, \dots, p-1$. Then, we get

$$\sum_{i=1}^n x_{ij} \psi \left(\frac{y_i - \mathbf{x}_i' \beta}{s} \right) = 0, \quad j = 0, 1, \dots, p-1,$$

where x_{ij} is the j th component of $\mathbf{x}_i$ and $\psi = \rho'$. Since ρ is nonlinear in β , we cannot get explicit solution. Instead, we use an iterative method, called IRLS (Iterative Reweighted Least Squares). Let

$$w_{i\beta} = \frac{\psi \{(y_i - \mathbf{x}_i' \beta) / s\}}{(y_i - \mathbf{x}_i' \beta) / s}, \quad i = 1, 2, \dots, n$$

and let $w_{i\beta} = 1$ if $y_i - \mathbf{x}_i' \beta = 0$. Then, we may write

$$\sum_{i=1}^n x_{ij} w_{i\beta} (y_i - \mathbf{x}_i' \beta) = 0, \quad j = 0, 1, \dots, p-1$$

and it is equivalent to

$$\sum_{i=1}^n x_{ij} w_{i\beta} \mathbf{x}_i' \beta = \sum_{i=1}^n x_{ij} w_{i\beta} y_i, \quad j = 0, 1, \dots, p-1.$$

In matrix notation, we have

$$\mathbf{X}'\mathbf{W}_\beta\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{W}_\beta\mathbf{y},$$

where $\mathbf{W}_\beta = \text{diag}(w_{1\beta}, w_{2\beta}, \dots, w_{n\beta})$. But, $\mathbf{W}_\beta$ contains unknown $\boldsymbol{\beta}$, $(\mathbf{X}'\mathbf{W}_\beta\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}_\beta\mathbf{y}$ cannot be a solution of $\boldsymbol{\beta}$. Let $\hat{\boldsymbol{\beta}}_0$ be an initial value of $\boldsymbol{\beta}$, and replace $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}_0$, then let $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'\mathbf{W}_{\hat{\boldsymbol{\beta}}_0}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}_{\hat{\boldsymbol{\beta}}_0}\mathbf{y}$. We continue this process, i.e.,

$$\hat{\boldsymbol{\beta}}_{q+1} = (\mathbf{X}'\mathbf{W}_{\hat{\boldsymbol{\beta}}_q}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}_{\hat{\boldsymbol{\beta}}_q}\mathbf{y}, \quad q = 0, 1, 2, \dots$$

which is called IRLS.

6.5.2 Influence function

If let $u = (y - \mathbf{x}'\boldsymbol{\beta})/s$, then $w(u) = \{\partial\rho(u)/\partial u\}/u$, $u = (y_i - \mathbf{x}'_i\boldsymbol{\beta})/s$, so that $\psi(u) = \partial\rho(u)/\partial u$, which is called *influence function*.

Table 6.8 (p. 264)

Fig. 6.11 (p. 265)

Ex. 6.8 (p. 265)

6.6 Inverse Regression

In the most cases of regression analysis, we predict response y at some $X = x$, however, we sometimes predict x_0 for a given response y_0 . This

problem is called *inverse regression* (*calibration, discrimination*).

Consider a simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

The estimate of x_0 for a given y_0 is

$$\hat{x}_0 = \frac{y_0 - \hat{\beta}_0}{\hat{\beta}_1}.$$

Also, the $100(1 - \alpha)\%$ C.I. for x_0 is obtained by solving

$$\frac{(y_0 - \hat{\beta}_0 - \hat{\beta}_1 x)^2}{s^2 A^2} \leq t^2,$$

where

$$A^2 = 1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\sum (x_i - \bar{x})^2}.$$

So, if we let $d = x - \bar{x}$, then the inequality becomes

$$d^2 \left\{ \hat{\beta}_1^2 - \frac{t^2 s^2}{\sum (x_i - \bar{x})^2} \right\} - 2d\hat{\beta}_1(y_0 - \bar{y}) + \left\{ (y_0 - \bar{y})^2 - t^2 s^2 \left(1 + \frac{1}{n}\right) \right\} = 0.$$

Let d_1, d_2 be solutions for d , and we have $X_L \leq X_0 \leq X_U$, where $X_L = \bar{X} + d_1$, $X_U = \bar{X} + d_2$.

Note that there are four types of solution for the 2nd degree polynomial inequality; (a, b) , $(-\infty, a)$, (b, ∞) , $(-\infty, \infty)$. Among them, only (a, b) is meaningful. Recall that the solution (a, b) can be obtained only

when $a_2 > 0$ in $a_2x^2 + a_1x + a_0 < 0$. Note that $a_2 > 0$ is equivalent to $\hat{\beta}_1^2 - \frac{t^2s^2}{\sum(x_i - \bar{x})} > 0$ which is the rejection region for $H_0 : \beta_1 = 0$.

Ex. 6.9 (p.268)

Chapter 7. Biased Estimation

7.1 James - Stein Shrinkage Method

(1) Motivation

Let $\mathbf{z} \sim N_p(\boldsymbol{\mu}, \mathbf{I})$, then $\mathbf{z}$ is a good estimator for $\boldsymbol{\mu}$ because it is MLE and UMVUE (uniformly minimum variance unbiased estimator). Even though $\mathbf{z}$ is a good estimator for $\boldsymbol{\mu}$ in the sense of 1st moment (i.e., $E(\mathbf{z}) = \boldsymbol{\mu}$), however, James and Stein (1961), considered the 2nd moment (i.e., MSE : mean squared error);

$$E(\mathbf{z}'\mathbf{z}) = \sum_{i=1}^p E(z_i^2) = \sum (1 + \mu_i^2) = p + \boldsymbol{\mu}'\boldsymbol{\mu} > \boldsymbol{\mu}'\boldsymbol{\mu}.$$

Hence, $\mathbf{z}'\mathbf{z}$ is not an unbiased estimator of $\boldsymbol{\mu}'\boldsymbol{\mu}$. Based on this idea, they considered

$$\tilde{\boldsymbol{\mu}} = c\mathbf{z}, \quad 0 < c < 1$$

where c is called *shrinkage* constant.

(2) James-Stein estimator

James and Stein showed that $\exists \quad 0 < c < 1$ s.t. $\text{MSE}(\tilde{\boldsymbol{\mu}}) \leq \text{MSE}(\mathbf{z})$. We will

prove it. Note that

$$\begin{aligned}
\text{MSE}(\tilde{\boldsymbol{\mu}}) &= E[(\tilde{\boldsymbol{\mu}} - \boldsymbol{\mu})'(\tilde{\boldsymbol{\mu}} - \boldsymbol{\mu})] \\
&= E[(\tilde{\boldsymbol{\mu}} - E(\tilde{\boldsymbol{\mu}}) + E(\tilde{\boldsymbol{\mu}}) - \boldsymbol{\mu})'(\tilde{\boldsymbol{\mu}} - E(\tilde{\boldsymbol{\mu}}) + E(\tilde{\boldsymbol{\mu}}) - \boldsymbol{\mu})] \\
&= E[(\tilde{\boldsymbol{\mu}} - E(\tilde{\boldsymbol{\mu}}))'(\tilde{\boldsymbol{\mu}} - E(\tilde{\boldsymbol{\mu}}))] + [E(\tilde{\boldsymbol{\mu}}) - \boldsymbol{\mu}]'[E(\tilde{\boldsymbol{\mu}}) - \boldsymbol{\mu}] \\
&= \text{tr}[\text{Cov}(\tilde{\boldsymbol{\mu}})] + ||\text{Bias}(\tilde{\boldsymbol{\mu}})||^2
\end{aligned}$$

and therefore,

$$\begin{aligned}
\text{MSE}(\tilde{\boldsymbol{\mu}}) &= \text{tr}(c^2 \mathbf{I}) + (\boldsymbol{\mu} - E(\tilde{\boldsymbol{\mu}}))'(\boldsymbol{\mu} - E(\tilde{\boldsymbol{\mu}})) \\
&= pc^2 + (1 - c)^2 \boldsymbol{\mu}' \boldsymbol{\mu}.
\end{aligned}$$

On the other hand,

$$\text{MSE}(\mathbf{z}) = p$$

To find c minimizing $\text{MSE}(\tilde{\boldsymbol{\mu}})$, we take 1st derivative w.r.t. c , i.e.,

$$0 = \frac{\partial \text{MSE}(\tilde{\boldsymbol{\mu}})}{\partial c} = 2cp - 2(1 - c)\boldsymbol{\mu}' \boldsymbol{\mu} = 2c(p + \boldsymbol{\mu}' \boldsymbol{\mu}) - 2\boldsymbol{\mu}' \boldsymbol{\mu}$$

which gives

$$c = \frac{\boldsymbol{\mu}' \boldsymbol{\mu}}{p + \boldsymbol{\mu}' \boldsymbol{\mu}}$$

and then, we have

$$\tilde{\boldsymbol{\mu}} = \left(1 - \frac{p}{p + \boldsymbol{\mu}' \boldsymbol{\mu}}\right) \mathbf{z}$$

But, this estimator contains unknown parameter $\mu'\mu$, we replace it by its unbiased estimator $z'z - p$, i.e.,

$$\tilde{\mu} = \left(1 - \frac{p}{z'z}\right) z$$

Finally, James and Stein showed that

$$\tilde{\mu}_s = \left[1 - \frac{(p-2)}{z'z}\right] z$$

gives the minimum MSE, and it is called *Stein shrinkage* estimator.

7.2 Ridge Regression

7.2.1 Inference on the ridge regression

(1) Motivation and definition

The LSE $\hat{\beta} = (X'X)^{-1}X'y$ will be very unstable if there exists multicollinearity. Hoerl and Kennard (1970) suggested the *ridge estimator* defined as

$$\hat{\beta}(\theta) = (X'X + \theta I)^{-1}X'y,$$

where θ is called *biasing parameter* or *shrinkage parameter*

(2) Estimation of θ

(a) ridge trace

First, we draw a figure of $\hat{\beta}_j(\theta)$, $j = 0, \dots, p - 1$ for $\theta > 0$, and we estimate θ where $\hat{\beta}_j(\theta)$ is stabilized. This method is called *ridge trace*.

Fig. 7.1 (p. 280)

(b) GCV_θ

Wahba, Golub, and Heath (1979) suggested minimizing the GCV (generalized cross validation), defined as

$$GCV_\theta = \sum e_{i,\theta}^2 / \left[1 - \frac{1}{n} \text{tr}(\mathbf{H}_\theta) \right]^2,$$

where $e_{i,\theta}$ is residual and $\mathbf{H}_\theta = \mathbf{X}(\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\mathbf{X}'$ is a hat matrix.

Ex.7.1 (p.280)

7.2.2 Properties of ridge estimator

(1) Regularized LSE

We can show that

$$\hat{\beta}(\theta) = \text{Arg}_{\beta} \min (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) \text{ s.t. } \beta'\beta = c^2$$

i.e.,

$$\hat{\beta}(\theta) = \text{Arg}_{\beta} \min (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) + \lambda(\beta'\beta - c^2),$$

where λ is a lagrangian multiplier.

Fig. 7.2 (p. 283)

(2) MSE

Theorem 7.1. $\text{MSE}(\hat{\beta}) - \text{MSE}(\hat{\beta}(\theta))$ is positive definite for some $\theta > 0$.

(Proof) Since

$$\begin{aligned} \text{MSE}(\hat{\beta}(\theta)) &= E[(\hat{\beta}(\theta) - \beta)(\hat{\beta}(\theta) - \beta)'] \\ &= \text{Cov}[(\hat{\beta}(\theta))] + [E(\hat{\beta}(\theta)) - \beta][E(\hat{\beta}(\theta)) - \beta]' \end{aligned}$$

and let $\mathbf{K} = (\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}$, then

$$\begin{aligned} \text{Cov}[\hat{\beta}(\theta)] &= \text{Cov}[(\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}] \\ &= (\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\sigma^2 \\ &= \mathbf{K}\mathbf{X}'\mathbf{X}\mathbf{K}\sigma^2 \end{aligned}$$

and

$$E(\hat{\beta}(\theta)) - \beta = E(\mathbf{K}\mathbf{X}'\mathbf{y}) - \beta = (\mathbf{K}\mathbf{X}'\mathbf{X} - \mathbf{K}\mathbf{K}^{-1})\beta = -\theta\mathbf{K}\beta$$

we have

$$\text{MSE}(\hat{\beta}(\theta)) = \mathbf{K}\mathbf{X}'\mathbf{X}\mathbf{K}\sigma^2 + \theta^2\mathbf{K}\beta\beta'\mathbf{K}$$

Here, $\text{Cov}(\hat{\beta}) = \text{MSE}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2$, so that

$$\begin{aligned} \text{MSE}(\hat{\beta}) - \text{MSE}(\hat{\beta}(\theta)) &= (\mathbf{X}'\mathbf{X})^{-1}\sigma^2 - \mathbf{K}\mathbf{X}'\mathbf{X}\mathbf{K}\sigma^2 - \theta^2\mathbf{K}\beta\beta'\mathbf{K} \\ &= \mathbf{K}\mathbf{K}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}^{-1}\mathbf{K}\sigma^2 - \mathbf{K}\mathbf{X}'\mathbf{X}\mathbf{K}\sigma^2 - \theta^2\mathbf{K}\beta\beta'\mathbf{K} \\ &= \mathbf{K}[\mathbf{K}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}^{-1}\sigma^2 - \mathbf{X}'\mathbf{X}\sigma^2 - \theta^2\beta\beta']\mathbf{K} \end{aligned}$$

Now, to show $\text{MSE}(\hat{\beta}) - \text{MSE}(\hat{\beta}(\theta))$ is p.d., we are only to show

$$\Delta = \mathbf{K}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}^{-1}\sigma^2 - \mathbf{X}'\mathbf{X}\sigma^2 - \theta^2\beta\beta'$$

is p.d. because $\mathbf{K}$ is p.d. Note that

$$\begin{aligned} \mathbf{K}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}^{-1} &= (\mathbf{X}'\mathbf{X} + \theta\mathbf{I})(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X} + \theta\mathbf{I}) \\ &= \mathbf{X}'\mathbf{X} + 2\theta\mathbf{I} + \theta^2(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

implies that

$$\Delta = \theta[\{2\mathbf{I} + \theta(\mathbf{X}'\mathbf{X})^{-1}\}\sigma^2 - \theta\beta\beta']$$

Now, to show Δ is p.d., we need to show $2\sigma^2\mathbf{I} - \theta\beta\beta'$ is p.d., because

$(\mathbf{X}'\mathbf{X})^{-1}$ is p.d and $\theta > 0$. Let $\mathbf{a} \neq \mathbf{0}$, then

$$\mathbf{a}'(2\sigma^2\mathbf{I} - \theta\beta\beta')\mathbf{a} = 2\sigma^2\mathbf{a}'\mathbf{a} - \theta(\mathbf{a}'\beta)^2 \geq 2\sigma^2\mathbf{a}'\mathbf{a} - \theta(\mathbf{a}'\mathbf{a})(\beta'\beta) = (2\sigma^2 - \theta\beta'\beta)\mathbf{a}'\mathbf{a}$$

by the Cauchy-Schwarz inequality. Hence, if $2\sigma^2 - \theta\beta'\beta > 0$, then $2\sigma^2\mathbf{I} - \theta\beta\beta'$ is p.d., i.e., For $0 < \theta < 2\sigma^2 / \beta'\beta$, $\text{MSE}(\hat{\beta}) - \text{MSE}(\hat{\beta}(\theta))$ is p.d..

(3) Bayes estimator

Consider

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2\mathbf{I})$$

and let

$$\beta \sim N\left(\mathbf{0}, \frac{\sigma^2}{\theta}\mathbf{I}\right)$$

be the prior distribution of β . Then, we can show that the posterior distribution of β is

$$N_p((\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}, (\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\sigma^2)$$

Therefore, under the squared error loss, $\hat{\beta}(\theta) = (\mathbf{X}'\mathbf{X} + \theta\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}$ is the Bayes estimator of β .

7.3 Principal Component Regression

(1) Motivation

When there are many covariates the PCR (principal component regression) uses a few principal components instead of the original covariates.

(1) Derivation of PCR

By the SVD (singular values decomposition) of the $n \times p$ matrix $\mathbf{X}$, we have

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}',$$

where $\mathbf{U}$ and $\mathbf{V}$ are $n \times p$ and $p \times p$ orthogonal matrix, respectively, and $\mathbf{D}$ is $p \times p$ diagonal matrix with diagonal elements $d_1 \geq d_2 \geq \cdots \geq d_p \geq 0$ which are singular values of $\mathbf{X}$. Therefore,

$$\mathbf{X}'\mathbf{X} = \mathbf{U}\mathbf{D}^2\mathbf{V}'$$

which is just the spectral decomposition of $\mathbf{X}'\mathbf{X}$. Here, $d_1^2 \geq d_2^2 \geq \cdots \geq d_p^2 \geq 0$ are eigenvalues of $\mathbf{X}'\mathbf{X}$ and the corresponding eigenvectors are $v_1, v_2, \cdots, v_p$ which are column vectors of $\mathbf{V}$, and they are orthogonal to each other. Let $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_p)$, where $\mathbf{x}_i$ is n -vector for the i th co-variate, and let

$$\mathbf{z}_i = \mathbf{X}v_i = \mathbf{u}_i d_i, \quad i = 1, 2, \cdots, p,$$

where $\mathbf{z}_i$ is called the i th principal component, and $\mathbf{u}_i$ is the i th column vector of $\mathbf{U}$. Hence, the sample variance of $\mathbf{z}_i$ is

$$\text{Var}(\mathbf{z}_i) = \text{Var}(\mathbf{X}v_i) = d_i^2/n, \quad i = 1, 2, \cdots, p.$$

Here, z_1 is called the 1st principal component, and the i -th principal component is $z_i = Xv_i$. Note that v_i is called the i th principal component direction, and it satisfies the following conditions;

$$\begin{aligned} & \max_{\|\alpha\|=1} \text{Var}(X\alpha), \\ & v_l' S \alpha = 0 \\ & l=1,2,\dots,i-1 \end{aligned}$$

where

$$S = X'X/n$$

is the sample variance-covariance matrix of covariates. Also, since $v_l' S \alpha = 0$, $z_i = X\alpha$ and $z_l = Xv_l$, $l = 1, 2, \dots, i-1$ are orthogonal to each other.

The PCR uses just few principals $z_1, z_2, \dots, z_M$, $M \ll p$ out of $z_1, z_2, \dots, z_p$.

If the response and covariates are centered, we may write the fitted vector of the PCR as

$$\hat{y} = \sum_{m=1}^M \hat{\theta}_m z_m,$$

where $\hat{\theta}_m = z_m' y / z_m' z_m$ which is just the estimate of the slope for the simple linear regression of y on z_m . To estimate M , the number of principals to be used, is often estimated by CV.

7.4 PLS : Partial Least Squares

PLS is quite similar to the PCR, however, it uses the information of $\mathbf{y}$ in addition to $\mathbf{X}$. Assume that the i th PLS direction vector $\mathbf{v}_i$ satisfies the following condition;

$$\begin{aligned} \max_{\substack{||\boldsymbol{\alpha}||=1 \\ \mathbf{v}_l' \mathbf{S} \boldsymbol{\alpha} = 0 \\ l=1,2,\dots,i-1}} \quad & \text{Corr}^2(\mathbf{y}, \mathbf{X}\boldsymbol{\alpha}) \text{Var}(\mathbf{X}\boldsymbol{\alpha}) \end{aligned}$$

The PLS algorithm can be summarized as follows;

(algorithm)

- (i) By regressing $\mathbf{y}$ on $\mathbf{x}_j, j = 1, 2, \dots, p$, compute the estimate of regression coefficients $\hat{v}_{1j} = \langle \mathbf{x}_j, \mathbf{y} \rangle$.
- (ii) Compute the 1st PLS direction vector $\mathbf{z}_1 = \sum \hat{v}_{1j} \mathbf{x}_j$.
- (iii) Regress $\mathbf{y}$ on $\mathbf{z}_1$, and compute the estimate of coefficient $\hat{\theta}_1 = \langle \mathbf{z}_1, \mathbf{y} \rangle$.
- (iv) Orthogonalize $\mathbf{x}_1, \dots, \mathbf{x}_p$ w.r.t. $\mathbf{z}_1$, i.e, $\mathbf{x}_j - \frac{\langle \mathbf{z}_1, \mathbf{x}_j \rangle}{\langle \mathbf{z}_1, \mathbf{z}_1 \rangle} \mathbf{z}_1, j = 1, \dots, p$.
- (v) Repeat this process to the M th direction.

7.5 LASSO

7.5.1 Estimation of LASSO

(1) Definition

LASSO (least absolute shrinkage and selection operator) is suggested by Tibshirani (1996). The ridge regression is obtained by assigning L_2 restriction on β , however, the LASSO is obtained by assigning L_1 restriction on β , i.e.,

$$\begin{aligned}\hat{\beta}_L &= \arg \min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) \text{ s.t. } \sum_{j=1}^p |\beta_j| \leq s \\ &= \arg \min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) + \lambda \sum_{j=1}^p |\beta_j|,\end{aligned}$$

where s and λ are parameters to be estimated. The solution of the above equation does not have a closed form solution, so that we are only to compute β numerically by packages like LARS (least angle regression).

We can obtain an explicit solution when we have one covariate. For simplicity, assume that $\sum y_i = 0$, $\sum x_i = 0$ and $\sum x_i^2 = 1$, and consider $y_i = \beta x_i + \epsilon_i$. Then, the LASSO estimate is given by

$$\hat{\beta}_L(\lambda) = \text{sgn}(\beta)(|\beta| - \lambda)_+,$$

where $(x)_+ = xI(x > 0)$. We can prove as follows; Note that

$$\begin{aligned}f(\beta) &= \frac{1}{2} \sum (y_i - x_i\beta)^2 + \lambda|\beta| \\ &= \frac{1}{2} \sum (y_i - x_i\hat{\beta} + x_i\hat{\beta} - x_i\beta)^2 + \lambda|\beta| \\ &= \frac{1}{2} \sum (y_i - x_i\hat{\beta})^2 + \frac{1}{2} (\hat{\beta} - \beta)^2 + \lambda|\beta|\end{aligned}$$

Hence,

$$\min_{\beta} f(\beta) \Leftrightarrow \min_{\beta} g(\beta) = \frac{1}{2}(\hat{\beta} - \beta)^2 + \lambda|\beta|$$

so that $\partial g(\beta)/\partial \beta = 0$ gives the LASSO estimate.

(2) Estimation of s .

Note that $\sum |\beta_j| \leq s \sum \beta_j^2 / |\beta_j| \leq s$, which is similar form to the ridge regression case. Hence, under this restriction, the ridge regression estimator is given by

$$\tilde{\beta} = (X'X + \lambda W^-)^{-1} Xy,$$

where $W = \text{diag}(|\tilde{\beta}_1|, \dots, |\tilde{\beta}_p|)$ and λ satisfies $\sum |\tilde{\beta}_j| = s$. Here, W^- denotes the generalized inverse of W which might contain $\tilde{\beta}_j = 0$. Now, the hat matrix is

$$H_s = X(X'X + \lambda W^-)^{-1} X'$$

and to estimate s , we minimize the generalized cross validation, i.e.,

$$GCV(s) = \frac{1}{n} \sum (y_i - \hat{y}_i)^2 / \{1 - \frac{1}{n} \text{tr}(H_s)\}^2$$

7.5.2 Properties of LASSO

(1) Geometric interpretation

Fig. 7.3 (p. 290)

(2) Bayesian interpretation

ridge regression estimator = posterior mean when the prior for β is Gaussian.

LASSO estimator = posterior mode when the prior for β is double exponential (Laplace).

(3) Comparisons

Under the orthonormal design, i.e., $X'X = I$,

(i) variable selection : $\hat{\beta}_{S,j} = \hat{\beta}_j I(|\hat{\beta}_j| > \lambda)$

(ii) ridge regression : $\hat{\beta}_{R,j} = \hat{\beta}_j / (1 + \lambda)$

(iii) LASSO : $\hat{\beta}_{L,j} = \text{sgn}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda)_+$

Fig. 7.4 (p. 290)

7.6 Example (Prostate Cancer Data)

(1) Data description

response : lpsa (log of the level of prostate specific antigen)

covariates : $X_1, \dots, X_8$

LSE, variable selection, ridge reg., LASSO, PCR, PLS : Table 7.4 (p. 293)

Fig. 7.7 and 7.8 (p. 294)

R code (Fig. 7.9, p. 294)

7.7 Bayes Estimator and Biased Estimators

7.7.1 Bayes Estimator

(1) Posterior distribution

Let $f(x; \theta)$ be the pdf of the r.v. X , i.e.,

$$X \sim f(x; \theta),$$

where θ is a unknown parameter. In the non-Bayesian point of view, θ is regarded as a unknown constant. On the other hand, in the Bayesian point of view, θ is regarded as a r.v. with pdf. Hence, in the Bayesian point of view, $f(x; \theta)$ is regarded as a conditional pdf of X when θ is given, i.e.,

$$X \sim f(x|\theta)$$

Now, let the pdf of θ be $g(\theta; \gamma)$, i.e.,

$$\theta \sim g(\theta; \gamma),$$

where γ is also a parameter. We call $g(\theta; \gamma)$, the pdf of θ , as a *prior distribution*.

Also, the product of the conditional pdf $f(x|\theta)$ and $g(\theta; \gamma)$, the prior pdf of θ , is called a *posterior distribution*, i.e.,

$$g(\theta|x; \gamma) = \frac{f(x|\theta)g(\theta; \gamma)}{\int f(x|\theta)g(\theta; \gamma)d\theta}$$

(2) Bayes estimator

Let $\delta \equiv \delta(x)$ be an estimator of θ , and let $L(\theta, \delta)$ be a loss function. Then, $L(\theta, \delta) = (\theta - \delta)^2$ is called a squared error loss function, and $L(\theta, \delta) = |\theta - \delta|$ is called an absolute error loss function. Further, the expectation of the loss function $R(\theta, \delta) = E[L(\theta, \delta)]$ is called a risk function, i.e.,

$$R(\theta, \delta) = \int_{\chi} L(\theta, \delta) f(x|\theta) dx$$

Now, the Bayes risk is defined as

$$r(\theta, \delta) = \int_{\Theta} R(\theta, \delta) g(\theta; \gamma) d\theta$$

which can be rewritten as

$$\begin{aligned} r(\theta, \delta) &= \int_{\Theta} R(\theta, \delta) g(\theta; \gamma) d\theta \\ &= \int_{\Theta} \int_{\chi} L(\theta, \delta) f(x|\theta) g(\theta; \gamma) dx d\theta \\ &= \int_{\chi} \int_{\Theta} L(\theta, \delta) f(x|\theta) g(\theta; \gamma) d\theta dx \\ &= \int_{\chi} E[L(\Theta, \delta) | X = x] dx, \end{aligned}$$

where $E[L(\Theta, \delta)|X = x]$ is called *posterior risk*. The *Bayes estimator* minimizes the Bayes risk or the posterior risk, i.e.,

$$\hat{\delta}_{Bayes} = \operatorname{argmin}_{\delta} r(\theta, \delta).$$

We can show that the Bayes estimator is a posterior mean under squared error loss, and a posterior mode under absolute error loss.

7.7.2 Stein shrinkage estimator and the empirical Bayes estimator

If the parameter γ in the prior pdf $g(\theta; \gamma)$ is unknown, it should be estimated. To estimate γ , we use the marginal pdf of the posterior pdf by integrating it w.r.t. θ . If we replace the estimate of γ in the Bayes estimator, the resulting estimator is called *empirical Bayes estimator*.

Theorem 7.2 Assume that $\mathbf{z} \sim N_p(\boldsymbol{\mu}, \mathbf{I})$ and the prior for $\boldsymbol{\mu}$ is $\boldsymbol{\mu} \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I})$. Then, the Stein shrinkage estimator is the empirical bayes estimator under the squared error loss.

(Proof) First, the posterior becomes

$$\begin{aligned}
& f(z|\mu)g(\mu) \\
& \propto \exp \left[-\frac{1}{2}(z - \mu)'(z - \mu) \right] \exp \left[-\frac{1}{2\sigma^2}\mu'\mu \right] \\
& \propto \exp \left[-\frac{1}{2}(z'z - 2\mu'z + \mu'\mu + \frac{1}{\sigma^2}\mu'\mu) \right] \\
& \propto \exp \left[-\frac{1}{2} \left\{ (1 + \frac{1}{\sigma^2})[\mu'\mu - 2\frac{\sigma^2}{1 + \sigma^2}\mu'z + \frac{\sigma^2}{1 + \sigma^2}z'z] \right\} \right].
\end{aligned}$$

If we let

$$w = \frac{\sigma^2}{1 + \sigma^2}$$

then, the posterior can be written as

$$\begin{aligned}
& \propto \exp \left[-\frac{1}{2w}(\mu'\mu - 2w\mu'z + w^2z'z) \right] \exp \left[-\frac{1}{2}(1 - w)z'z \right] \\
& \propto \exp \left[-\frac{1}{2w}(\mu - wz)'(\mu - wz) \right] \exp \left[-\frac{1}{2}(1 - w)z'z \right].
\end{aligned}$$

Therefore, the posterior becomes

$$\mu|z \sim N_p(wz, wI).$$

Since the Bayes estimator is the posterior mean under squared error loss,

we have

$$wz = \frac{\sigma^2}{1 + \sigma^2}z = \left(1 - \frac{1}{1 + \sigma^2}\right)z,$$

however, it contains unknown parameter σ^2 , we obtain the marginal distribution of $\mathbf{z}$ to compute the estimator of σ^2 .

$$\int f(\mathbf{z}|\boldsymbol{\mu})g(\boldsymbol{\mu})d\boldsymbol{\mu} \propto \exp[-\frac{1}{2}(1-w)\mathbf{z}'\mathbf{z}], \mathbf{z} \sim N_p(\mathbf{0}, (1+\sigma^2)\mathbf{I})$$

Now, consider

$$V \equiv \frac{\mathbf{z}'\mathbf{z}}{1+\sigma^2} \sim \chi^2(p)$$

then

$$E\left(\frac{1}{V}\right) = \int_0^\infty \frac{1}{v} \frac{v^{\frac{p}{2}-1} e^{-v/2}}{\Gamma(\frac{p}{2}) 2^{\frac{p}{2}}} dv = \frac{\Gamma(\frac{p}{2}-1) 2^{\frac{p}{2}-1}}{\Gamma(\frac{p}{2}) 2^{\frac{p}{2}}} = \frac{\Gamma(\frac{p}{2}-1)}{2(\frac{p}{2}-1)\Gamma(\frac{p}{2}-1)} = \frac{1}{p-2}$$

hence,

$$E\left[\frac{1+\sigma^2}{\mathbf{z}'\mathbf{z}}\right] = p-2$$

Therefore, the unbiased estimator of $(1+\sigma^2)^{-1}$ is $(p-2)/\mathbf{z}'\mathbf{z}$, and finally the empirical Bayes estimator is

$$\left(1 - \frac{p-2}{\mathbf{z}'\mathbf{z}}\right)\mathbf{z}$$

which is just the Stein shrinkage estimator.

7.7.3 Bayes estimator in regression

Consider a multiple linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

which is equivalent to $\mathbf{y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, and assume that the prior to $\boldsymbol{\beta}$ be

$$\boldsymbol{\beta} \sim N_p(\mathbf{m}, \sigma^2\mathbf{V})$$

Now, the posterior of $\boldsymbol{\beta}$ becomes

$$\begin{aligned} & \exp \left[-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right] \exp \left[-\frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{m})'\mathbf{V}^{-1}(\boldsymbol{\beta} - \mathbf{m}) \right] \\ & \propto \exp \left[-\frac{1}{2\sigma^2}(\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{V}^{-1}\boldsymbol{\beta} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{V}^{-1}\mathbf{m} + \dots) \right] \\ & \propto \exp \left[-\frac{1}{2\sigma^2}\{\boldsymbol{\beta}'(\mathbf{X}'\mathbf{X} + \mathbf{V}^{-1})\boldsymbol{\beta} - 2\boldsymbol{\beta}'(\mathbf{V}^{-1}\mathbf{m} + \mathbf{X}'\mathbf{y}) + \dots\} \right] \\ & \equiv \exp \left[-\frac{1}{2\sigma^2}\{(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta)'(\mathbf{X}'\mathbf{X} + \mathbf{V}^{-1})(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta)\} \right], \end{aligned}$$

where $\boldsymbol{\mu}_\beta$ is the mean of the posterior, then we let

$$-2\boldsymbol{\beta}'(\mathbf{X}'\mathbf{X} + \mathbf{V}^{-1})\boldsymbol{\mu}_\beta \equiv -2\boldsymbol{\beta}'(\mathbf{V}^{-1}\mathbf{m} + \mathbf{X}'\mathbf{y})$$

Hence,

$$\boldsymbol{\mu}_\beta = (\mathbf{X}'\mathbf{X} + \mathbf{V}^{-1})^{-1}(\mathbf{V}^{-1}\mathbf{m} + \mathbf{X}'\mathbf{y}) \equiv \tilde{\boldsymbol{\beta}}_{Bayes}$$

which is the Bayes estimator under the squared error loss.

As a special case, let $\mathbf{V} = \lambda^{-1}\mathbf{I}_p$ for some $\lambda > 0$ and $\mathbf{m} = \mathbf{0}$, then

$$\tilde{\boldsymbol{\beta}}_{Bayes} = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}$$

which is just the ridge regression estimator. Next, if $\mathbf{V} = \lambda^{-1}(\mathbf{X}'\mathbf{X})^{-1}$ and $\mathbf{m} = \mathbf{0}$, then

$$\tilde{\boldsymbol{\beta}}_{Bayes} = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (1 + \lambda)^{-1}\hat{\boldsymbol{\beta}}$$

which is called *James – Stein regression estimator*.

Chapter 8. Generalized linear model (I)

8.1 Data and Statistical Models

(1) Types of data

data : numerical (quantitative) and categorical (qualitative)

- numerical : continuous and discrete
- categorical : nominal and ordinal

(2) Types of categorical data

- response : binary (dichotomous) and polytomous (polychotomous)
- categorical covariate : factor and level

(3) Types of models

- categorical response : logistic and log-linear
- continuous response : multiple linear regression and ANOVA

8.2 Exponential Family

(1) Definition

Let Y be r.v. with pdf

$$f(y; \theta, \phi) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right],$$

then Y belongs to an *exponential family* with *natural (canonical)* parameter θ if ϕ is known. Also, we assume that $a(\cdot)$, $b(\cdot)$, $c(\cdot, \cdot)$ are known functions.

(i) $N(\mu, \sigma^2)$ case

$$\begin{aligned} f(y; \theta, \phi) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\} \\ &= \exp \left\{ \frac{1}{\sigma^2} \left(y\mu - \frac{\mu^2}{2} \right) - \frac{1}{2} \left(\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2) \right) \right\} \end{aligned}$$

so that $\theta = \mu$, $\phi = \sigma^2$ and

$$a(\phi) = \phi, \quad b(\theta) = \frac{\theta^2}{2}, \quad c(y, \phi) = -\frac{1}{2} \left\{ \frac{y^2}{\sigma^2} + \log(2\pi\sigma^2) \right\}$$

(ii) $P(\lambda)$ case

$$\begin{aligned} f(y; \theta, \phi) &= \lambda^y e^{-\lambda} y!, \quad y = 0, 1, 2, \dots \\ &= \exp(y \log \lambda - \lambda - \log y!) \end{aligned}$$

so that $\theta = \log \lambda$, $\phi = 1$ and

$$a(\phi) = 1, \quad b(\theta) = e^\theta, \quad c(y, \phi) = -\log y!$$

(iii) $B(n, \pi)$ case

$$\begin{aligned} f(y; \theta, \phi) &= \binom{n}{y} \pi^y (1 - \pi)^{n-y}, \quad y = 0, 1, \dots, n \\ &= \exp \left\{ \frac{\frac{1}{n} y \log \left(\frac{\pi}{1 - \pi} \right) + \log(1 - \pi)}{1/n} + \log \binom{n}{y} \right\} \end{aligned}$$

so that $\theta = \log \left(\frac{\pi}{1 - \pi} \right)$, $\phi = 1/n$ and

$$a(\phi) = \phi, \quad b(\theta) = \log(1 + e^\theta), \quad c(y, \phi) = \log \binom{1/\phi}{y}$$

(2) Properties

Let

$$l(\theta; \phi, y) = \log f(y; \theta, \phi)$$

be the log-likelihood function, then we have the following theorem, called *Bartlett identity*.

Theorem 8.1

$$E \left(\frac{\partial l}{\partial \theta} \right) = 0, \quad E \left(\frac{\partial^2 l}{\partial \theta^2} \right) + E \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right] = 0$$

(Proof)

$$E \left(\frac{\partial l}{\partial \theta} \right) = \int \frac{1}{f(y; \theta)} \left(\frac{\partial}{\partial \theta} \right) f(y; \theta) dy = \int \frac{\partial}{\partial \theta} f(y; \theta) dy = \frac{\partial}{\partial \theta} \int f(y; \theta) dy = 0$$

$$\frac{\partial^2 l}{\partial \theta^2} = \frac{\partial}{\partial \theta} \left(\frac{\frac{\partial}{\partial \theta} f(y; \theta)}{f(y; \theta)} \right) = \frac{\frac{\partial^2}{\partial \theta^2} f(y; \theta)}{f(y; \theta)} - \left\{ \frac{\frac{\partial}{\partial \theta} f(y; \theta)}{f(y; \theta)} \right\}^2$$

$$E \left(\frac{\partial^2 l}{\partial \theta^2} \right) = \int \frac{\partial^2}{\partial \theta^2} f(y; \theta) dy - E \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right] = \frac{\partial^2}{\partial \theta^2} \int f(y; \theta) dy - E \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right] = -E \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right]$$

which completes the proof.

We call $U = \frac{\partial l}{\partial \theta}$ *score function* and

$$-E \left(\frac{\partial^2 l}{\partial \theta^2} \right) = E \left[\left(\frac{\partial l}{\partial \theta} \right)^2 \right]$$

is called *Fisher information number*. Now, we apply the Bartlett identity in Theorem 8.1 to the exponential family. The log-likelihood function is

$$l(\theta; y) = \frac{\{y\theta - b(\theta)\}}{a(\phi)} + c(y, \phi)$$

and we have

$$\frac{\partial l}{\partial \theta} = \frac{\{y - b'(\theta)\}}{a(\phi)}, \quad \frac{\partial^2 l}{\partial \theta^2} = -\frac{b''(\theta)}{a(\phi)}$$

hence, we have

$$E(Y) = \mu = b'(\theta).$$

Also, from $-\frac{b''(\theta)}{a(\phi)} + \frac{Var(Y)}{a^2(\phi)} = 0$, we have

$$Var(Y) = b''(\theta)a(\phi).$$

We call $b''(\theta)$ *variance function*, and ϕ *dispersion parameter*. Further, we can express θ in terms of $\mu = E(Y)$, and we denote the variance function as $V(\mu)$.

8.3 Construction of GLMs

The GLM (*Generalized Linear Models*), suggested by Nelder and Wedderburn (1972), consists of 3 parts;

1. $Y_1, \dots, Y_n$ are independent and belongs to an exponential family.
2. $\eta = \sum_{j=0}^{p-1} X_j \beta_j$ is called *linear predictor*, where $X_0 \equiv 1$.
3. There exists a function g , called a *link function*, s.t. $g(\mu_i) = \eta_i$, where η_i is the linear predictor and $\mu_i = E(Y_i)$. Also, g is assumed to be monotone and differentiable

The classical multiple linear model can be regarded as a special case of GLM with

$$\mu_i = \sum_{j=0}^{p-1} x_{ij} \beta_j = \eta_i$$

with identity link function.

For the binomial response, 3 popular link functions are as follows;

1. logit : $\eta = \log\{\mu/(1 - \mu)\}$, $0 < \mu < 1$
2. probit : $\eta = \Phi^{-1}(\mu)$, $\Phi(\cdot)$
3. complementary log-log : $\eta = \log\{-\log(1 - \mu)\}$

A link function satisfying $\theta = \eta$ is called *natural (canonical) link*, and natural link functions for each distribution is as follows;

normal	: $\eta = \mu$
Poisson	: $\eta = \log\mu$
binomial	: $\eta = \log\{\mu/(1 - \mu)\}$
gamma	: $\eta = \mu^{-1}$
inverse Gaussian	: $\eta = \mu^{-2}$

Also, properties of exponential families are summarized in the followings;

distribution	Normal	Poisson	Binomial	Gamma	Inverse Gaussian
notation	$N(\mu, \sigma^2)$	$P(\mu)$	$B(m, \pi)$	$G(\mu, \nu)$	$IG(\mu, \sigma^2)$
dispersion	σ^2	1	$1/m$	$\nu\mu^{-1}$	σ^2
$b(\theta)$	$\theta^2/2$	e^θ	$\log(1 + e^\theta)$	$-\log(-\theta)$	$-(-2\theta)^{1/2}$
$\mu(\theta)$	θ	e^θ	$e^\theta/(1 + e^\theta)$	$-1/\theta$	$(-2\theta)^{-1/2}$
natural link	identity	log	logit	inverse	$1/\mu^2$
$V(\mu)$	1	μ	$\mu(1 - \mu)$	μ^2	μ^3

8.4 Estimation of Regression Coefficients

In the GLMs, estimation of β is done by the maximum likelihood method. But, the score function

$$\frac{\partial l}{\partial \beta} = \left(\frac{\partial l}{\partial \beta_j} \right)_{j=0, \dots, p-1} = \mathbf{0}$$

does not give an explicit solution for β . Therefore, we use iteration methods such as *Newton – Raphson method* or *Fisher's scoring method*. In fact, they are based on the Taylor expansion. The 1st Taylor expansion of $f(y)$ about $y = \mu$ is

$$f(y) \simeq f(\mu) + (y - \mu)f'(\mu)$$

and we apply it to $\partial l / \partial \beta_j$, then we have

$$\begin{aligned} 0 &= \frac{\partial l}{\partial \beta_j} \\ &\simeq \left. \frac{\partial l}{\partial \beta_j} \right]_{\beta_j = \hat{\beta}_j^{(0)}} + (\beta_j - \hat{\beta}_j^{(0)}) \cdot \left. \frac{\partial^2 l}{\partial \beta_j^2} \right]_{\beta_j = \hat{\beta}_j^{(0)}}, \quad j = 0, \dots, p-1, \end{aligned}$$

where $\hat{\beta}_j^{(0)}$ is an initial value. In matrix notation for $\beta_j \quad j = 0, 1, \dots, p-1$,

$$\begin{aligned} \mathbf{0} &= \frac{\partial l}{\partial \beta} \\ &\simeq \left. \frac{\partial l}{\partial \beta} \right]_{\beta = \hat{\beta}^{(0)}} + \left. \frac{\partial^2 l}{\partial \beta \partial \beta'} \right]_{\beta = \hat{\beta}^{(0)}} \cdot (\beta - \hat{\beta}^{(0)}) \end{aligned}$$

and let the solution of β in the above equation be $\hat{\beta}^{(1)}$, then

$$\hat{\beta}^{(1)} = \hat{\beta}^{(0)} - A^{-1}(\hat{\beta}^{(0)}) \cdot \mathbf{u}(\hat{\beta}^{(0)}),$$

where $A^{-1}(\hat{\beta}^{(0)})$ is the inverse matrix of $p \times p$ matrix

$$A(\hat{\beta}^{(0)}) = - \left[\frac{\partial^2 l}{\partial \beta \partial \beta'} \Big|_{\beta = \hat{\beta}^{(0)}} \right]$$

and $\mathbf{u}(\hat{\beta}^{(0)})$ is given by

$$\mathbf{u}(\hat{\beta}^{(0)}) = \left[\frac{\partial l}{\partial \beta} \Big|_{\beta = \hat{\beta}^{(0)}} \right]$$

Next, we iterate the above process using $\hat{\beta}^{(1)}$ as an initial vector, and we have

$$\hat{\beta}^{(t+1)} = \hat{\beta}^{(t)} + A^{-1}(\hat{\beta}^{(t)}) \cdot \mathbf{u}(\hat{\beta}^{(t)}), \quad t = 0, 1, \dots$$

We continue this process until $\hat{\beta}^{(t+1)}$ is close enough to $\hat{\beta}^{(t)}$. If we use

$$I = -E \left(\frac{\partial^2 l}{\partial \beta \partial \beta'} \right)$$

instead of A , then it is called Fisher's scoring method.

Theorem 8.2 We can express

$$\mathbf{u} = \mathbf{X}'\mathbf{W}(\mathbf{y} - \boldsymbol{\mu}) \frac{\partial \eta}{\partial \boldsymbol{\mu}},$$

where $\mathbf{W} = \text{diag}(w_{ii}) = \mathbf{V}^{-1} \left(\frac{\partial \mu}{\partial \eta} \right)^2$ and $\mathbf{V} = a(\phi) \text{diag}(v_{ii})$. Also,

$$\mathbf{I} = \mathbf{X}' \mathbf{W} \mathbf{X}$$

(Proof) First, we compute $\mathbf{U} = (U_0, \dots, U_{p-1})'$, where

$$U_j = \frac{\partial l}{\partial \beta_j} = \sum \frac{\partial l_i}{\partial \beta_j} = \sum \frac{\partial l_i}{\partial \theta} \frac{\partial \theta}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j}$$

and

$$l = \sum l_i = \sum [y_i \theta - b_i(\theta) a(\phi) + c(y_i, \phi)] = \frac{1}{a(\phi)} \sum \{y_i \theta - b(\theta)\} + \sum c(y_i, \phi)$$

and hence

$$\begin{aligned} \frac{\partial l_i}{\partial \theta} &= \frac{1}{a(\phi)} (y_i - b'(\theta)) = \frac{1}{a(\phi)} (y_i - \mu_i) \\ \frac{\partial \theta}{\partial \mu_i} &= \left(\frac{\partial \mu_i}{\partial \theta} \right)^{-1} = \left(\frac{\partial b_i'(\theta)}{\partial \theta} \right)^{-1} = \frac{1}{v_{ii}} \\ \frac{\partial \eta_i}{\partial \beta_j} &= x_{ij} \end{aligned}$$

so that

$$U_j = \sum_i \frac{(y_i - \mu_i)}{a(\phi)} \frac{1}{v_{ii}} \frac{\partial \mu_i}{\partial \eta_i} x_{ij} = \sum_i (y_i - \mu_i) w_{ii} \frac{\partial \eta_i}{\partial \mu_i} x_{ij},$$

where $w_{ii} = \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 / a(\phi) v_{ii}$.

On the other hand, we compute $I = (I_{jk})$, $j, k = 0, \dots, p-1$, then

$$\begin{aligned} I_{jk} &= -E \left(\frac{\partial^2 l}{\partial \beta_j \partial \beta_k} \right) = -E \left(\frac{\partial U_j}{\partial \beta_k} \right) \\ &= -E \left[\frac{\partial}{\partial \beta_k} \left\{ \sum (y_i - \mu_i) w_{ii} \frac{\partial \eta_i}{\partial \mu_i} x_{ij} \right\} \right] = \sum \frac{\partial \mu_i}{\partial \beta_k} w_{ii} \frac{\partial \eta_i}{\partial \mu_i} x_{ij} \\ &= \sum \frac{\partial \eta_i}{\partial \beta_k} w_{ii} x_{ij} = \sum x_{ik} w_{ii} x_{ij} \end{aligned}$$

which completes the proof.

Now, the Fisher's scoring method is

$$\hat{\beta}^{(t+1)} = \hat{\beta}^{(t)} + I^{-1}(\hat{\beta}^{(t)}) U(\hat{\beta}^{(t)})$$

and if we multiply $I(\hat{\beta}^{(t)})$ on both sides, we have

$$I(\hat{\beta}^{(t)}) \hat{\beta}^{(t+1)} = I(\hat{\beta}^{(t)}) \hat{\beta}^{(t)} + U(\hat{\beta}^{(t)})$$

Now, if we use the result of Theorem 8.2, then the left hand side becomes

$$X' W X \hat{\beta}^{(t+1)}$$

and the right hand side becomes

$$X' W X \hat{\beta}^{(t)} + X' W (y - \hat{\mu}) \frac{\partial \eta}{\partial \mu} = X' W \left\{ \hat{\eta}^{(t)} + (y - \hat{\mu}^{(t)}) \frac{\partial \eta}{\partial \mu} \right\} = X' W z^{(t)},$$

where $\hat{\eta}^{(t)} = X \hat{\beta}^{(t)}$, $\hat{\mu}^{(t)} = g^{-1}(\hat{\eta}^{(t)})$ and

$$z^{(t)} = \hat{\eta}^{(t)} + (y - \hat{\mu}^{(t)}) \frac{\partial \eta}{\partial \mu}$$

which is called *adjusted dependent variable*. Finally, we have

$$\mathbf{X}'\mathbf{W}\mathbf{X}\hat{\boldsymbol{\beta}}^{(t+1)} = \mathbf{X}'\mathbf{W}\mathbf{z}^{(t)}, \quad t = 0, 1, \dots$$

and it has the same form as the weighted LSE, so that it is called *IRLS* (*Iterative Reweighted Least Squares*).

8.5 Goodness-of-Fit Measures for GLMs

There are two types of goodness-of-fit measures for GLMs : deviance and Pearson's chi-square statistic

(1) Deviance

Let $\hat{\boldsymbol{\beta}}_{max}$ and $\hat{\boldsymbol{\beta}}$ be estimators of $\boldsymbol{\beta}$ under the maximal model and the current model, respectively. Also, let L be the likelihood function. Then, the likelihood ratio test statistic is

$$\lambda = \frac{L(\hat{\boldsymbol{\beta}}_{max}; y)}{L(\hat{\boldsymbol{\beta}}; y)}.$$

If the current model is good, then λ will be close to 1, and if not, then it will be very large. Based on this idea, Nelder and Wedderburn (1972) suggested the *deviance*, defined as

$$D = 2 \log \lambda = 2[l(\hat{\boldsymbol{\beta}}_{max}; y) - l(\hat{\boldsymbol{\beta}}; y)]$$

It can be shown that the deviance D is asymptotically χ^2 distribution with d.f. $n - p$ if the model is good (i.e., under the null hypothesis). Hence, we conclude (reject the null hypothesis) that the current model is not good if $D > \chi^2_\alpha(n - p)$.

Ex.8.5 (p.313) (Normal dist.) Let $Y_1, \dots, Y_n$ be independent $N(\mu_i, \sigma^2)$, then the log-likelihood becomes

$$l(\boldsymbol{\beta}; \mathbf{y}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_i)^2 - \frac{n}{2} \log(2\pi\sigma^2)$$

Now, under the maximal model, we have $E(Y_i) = \mu_i$, $i = 1, \dots, n$, then the MLE of μ_i is $\hat{\mu}_i = y_i$. Therefore, the log-likelihood under the maximal model is

$$l(\hat{\boldsymbol{\beta}}_{max}; \mathbf{y}) = -\frac{n}{2} \log(2\pi\sigma^2)$$

On the other hand, if we assume that the current model is $E(Y_i) = \mu$ (i.e., one parameter), then the MLE of μ is $\hat{\mu} = \bar{y}$. Hence, the log-likelihood under the current model becomes

$$l(\hat{\boldsymbol{\beta}}; \mathbf{y}) = -\frac{1}{2\sigma^2} \sum (y_i - \bar{y})^2 - \frac{n}{2} \log(2\pi\sigma^2).$$

Therefore, the deviance is

$$D = 2 \left[l(\hat{\boldsymbol{\beta}}_{max}; \mathbf{y}) - l(\hat{\boldsymbol{\beta}}; \mathbf{y}) \right] = \frac{1}{\sigma^2} \sum (y_i - \bar{y})^2$$

which follows exactly χ^2 distribution with d.f. $n - 1$.

Ex.8.6 (p.314) (Poisson dist.) Let $Y_1, \dots, Y_n$ be independent $P(\lambda_i)$, then the log-likelihood function is

$$l(\boldsymbol{\beta}; \mathbf{y}) = \sum y_i \log \lambda_i - \sum \lambda_i - \sum \log y_i!$$

and under the maximal model $E(Y_i) = \lambda_i$, the MLE is $\hat{\lambda}_i = y_i$, so that

$$l(\hat{\boldsymbol{\beta}}_{max}; \mathbf{y}) = \sum y_i \log y_i - \sum y_i - \sum \log y_i!$$

On the other hand, if we assume that the current model is $E(Y_i) = \lambda$ (i.e., one parameter), then the MLE of λ is $\hat{\lambda} = \bar{y}$. Hence, the log-likelihood under the current model becomes

$$l(\hat{\boldsymbol{\beta}}; \mathbf{y}) = \sum y_i \log \bar{y} - n\bar{y} - \sum \log y_i!$$

Hence, the deviance is

$$D = 2 \left[\sum y_i \log y_i - \sum y_i \log \bar{y} \right] = 2 \sum y_i \log(y_i / \bar{y})$$

which follows approximately $\chi^2(n - 1)$.

For the normal distribution with p parameters $\beta_0, \beta_1, \dots, \beta_{p-1}$ in the current model, let the MLE of μ_i , $i = 1, \dots, n$ be $\hat{\mu}_i$, then the log-likelihood is

$$l(\hat{\boldsymbol{\beta}}; \mathbf{y}) = -\frac{1}{2\sigma^2} \sum (y_i - \hat{\mu}_i)^2 - \frac{n}{2} \log(2\pi\sigma^2)$$

and, therefore, the deviance is

$$D = 2[l(\hat{\beta}_{max}; \mathbf{y}) - l(\hat{\beta}; \mathbf{y})] = \sum (y_i - \hat{\mu}_i)^2 / \sigma^2$$

which is exactly the same as SSE / σ^2 in the multiple regression model, and follows $\chi^2(n - p)$.

Here are deviance for the distributions in the exponential family.

$$\text{Normal} : \sum (y_i - \hat{\mu}_i)^2 / \sigma^2$$

$$\text{Poisson} : 2 \sum \{ y_i \log(y_i / \hat{\mu}_i) - (y_i - \hat{\mu}_i) \}$$

$$\text{Binomial} : 2 \sum \{ y_i \log(y_i / \hat{\mu}_i) + (m_i - y_i) \log[(m_i - y_i) / (m_i - \hat{\mu}_i)] \}$$

$$\text{Gamma} : 2 \sum \{ -\log(y_i / \hat{\mu}_i) + (y_i - \hat{\mu}_i) / \hat{\mu}_i \}$$

$$\text{Inverse gaussian} : \sum (y_i - \hat{\mu}_i)^2 / (\hat{\mu}_i^2 y_i)$$

(2) Pearson's χ^2 statistic

Another goodness-of-fit measure in the GLMs is the Pearson's χ^2 defined as

$$X^2 = \sum (y_i - \hat{\mu}_i)^2 / V(\hat{\mu}_i)$$

where $V(\hat{\mu}_i)$ is the variance function. It can be shown that X^2 is asymptotically $\chi^2(n - p)$. Under the Gaussian distribution, $D = X^2$ and X^2 follows exactly $\chi^2(n - p)$.

8.6 Testing and Residuals

(1) Testing

Assume that the current model consists of p -parameters $\boldsymbol{\beta} = (\beta_0, \dots, \beta_{q-1}, \beta_q, \dots, \beta_{p-1})'$, and consider

$$H_0 : \beta_q = \dots = \beta_{p-1} = 0$$

Hence, under H_0 , the model consists of q parameters. Now, the test statistics is the difference between (D_1) (deviance under the current model) and (D_0) (deviance under the null model) Let $\hat{\boldsymbol{\beta}}_0$ and $\hat{\boldsymbol{\beta}}_1$ be estimates of $\boldsymbol{\beta}$ under the null model and the current model, respectively. Then,

$$\begin{aligned}\Delta D &= D_0 - D_1 \\ &= 2[l(\hat{\boldsymbol{\beta}}_{max}; \mathbf{y}) - l(\hat{\boldsymbol{\beta}}_0; \mathbf{y})] - 2[l(\hat{\boldsymbol{\beta}}_{max}; \mathbf{y}) - l(\hat{\boldsymbol{\beta}}_1; \mathbf{y})] \\ &= 2[l(\hat{\boldsymbol{\beta}}_1; \mathbf{y}) - l(\hat{\boldsymbol{\beta}}_0; \mathbf{y})].\end{aligned}$$

Since D_1 is asymptotically $\chi^2(n - p)$, and D_0 is asymptotically $\chi^2(n - q)$, it can be shown that ΔD is asymptotically $\chi^2(p - q)$. Therefore, we reject H_0 if $\Delta D > \chi^2_{\alpha}(p - q)$.

(2) Residuals

(i) Pearson residual

The Pearson residual is defined as

$$r_{Pi} = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

and note that

$$X^2 = \sum_{i=1}^n r_{Pi}^2$$

(ii) Anscombe residual

Since the Pearson residual do not follow the normal distribution, Anscombe suggested a transformation $A(\cdot)$, defined as

$$A(\cdot) = \int V^{-1/3}(\mu) d\mu$$

For example, in the Poisson distribution, $V(\mu) = \mu$, so that

$$\int \mu^{-1/3} d\mu = \frac{3}{2} \mu^{2/3}$$

and by the delta method, we have $\sqrt{V(A(Y))} \approx A'(\mu) \sqrt{V(\mu)} = \mu^{-1/3} \mu^{1/2} = \mu^{1/6}$. Hence,

$$r_{Ai} = \frac{\frac{3}{2}(y_i^{2/3} - \hat{\mu}_i^{2/3})}{\hat{\mu}_i^{1/6}}$$

(iii) deviance residual

Recall that $D = \sum d_i$, and the deviance residual is defined as

$$r_{Di} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

For example, in the Poisson case,

$$r_{Di} = \text{sign}(y_i - \hat{\mu}_i) \left\{ 2(y_i \log \left(\frac{y_i}{\hat{\mu}_i} \right) - y_i + \hat{\mu}_i) \right\}^{1/2}$$

8.7 ANOVA models

(1) ANOVA models

analysis of variance model : continuous response and categorical covariates

one covariate case : one-factor experiment (one-way classification)

two covariates case : two-factor experiment (two-way classification)

(2) Estimation

We can write the ANOVA model as a multiple linear regression model

$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, however, $\mathbf{X}$ is not a full-rank matrix, so that $\mathbf{X}'\mathbf{X}$ is singular.

Therefore, we cannot have the unique solution for $\boldsymbol{\beta}$ in the normal equation;

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$$

In this case, we have to assign some constraints. Assume that $\mathbf{X}$ is $n \times p$ ($p < n$) matrix. If q column vectors are linearly independent among p column vectors, then we have to assign $p - q$ constraints to $\boldsymbol{\beta}$. There are two methods of assigning constraints; *sum-to-zero constraint* and *corner-point constraint*.

First, we consider oneway ANOVA, and data structure for oneway ANOVA with J levels are given in Table 8.5.

Data structure for oneway ANOVA with J levels					
level	response				total
A_1	Y_{11}	Y_{12}	$\cdots$	Y_{1n_1}	$Y_{1\cdot}$
A_2	Y_{21}	Y_{22}	$\cdots$	Y_{2n_2}	$Y_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
A_J	Y_{J1}	Y_{J2}	$\cdots$	Y_{Jn_J}	$Y_{J\cdot}$

We assume that $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$, and let $N = \sum_{j=1}^J n_j$ and

$$\mathbf{y} = (Y_{11}, \cdots, Y_{n_1}, \cdots, Y_{J1}, \cdots, Y_{Jn_J})'$$

Further, we assume that $n_j \equiv K$ for all $j = 1, \cdots, J$, so that $N = JK$.

Also, note that the deviance can be written as

$$D = \frac{1}{\sigma^2} \sum (y_i - \hat{y}_i)^2 = \frac{1}{\sigma^2} (\mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}' \mathbf{X}'\mathbf{y})$$

(i) sum-to-zero constraint

Note that we may write

$$E(Y_{jk}) = \mu + \alpha_j, \quad j = 1, \dots, J; \quad k = 1, \dots, K,$$

where μ is an overall effect of the response and α_j is the relative effect of j th level. When we write $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, then

$$\mathbf{X}_{JK \times (J+1)} = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{1} & \mathbf{0} & \mathbf{1} & \dots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{1} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{1} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \mu \\ \alpha_1 \\ \vdots \\ \alpha_J \end{bmatrix}$$

where $\mathbf{1} = (1, \dots, 1)'$ and $\mathbf{0} = (0, \dots, 0)'$ are K -vectors. Hence,

$$\mathbf{X}'\mathbf{X}_{(J+1) \times (J+1)} = \begin{bmatrix} N & K & \dots & K \\ K & K & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ K & 0 & 0 & \dots & K \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} Y_{..} \\ Y_{1.} \\ \vdots \\ Y_{J.} \end{bmatrix}$$

Note that the 1st column of $\mathbf{X}'\mathbf{X}$ is sum of the other J columns, so that the rank of $\mathbf{X}'\mathbf{X}$ is J , and therefore, we need one constraint on $\boldsymbol{\beta}$. Here, we assign sum-to-zero-constraint, i.e., $\sum_{j=1}^J \alpha_j = 0$. Under this constraint,

$$\hat{\mu} = \frac{Y_{..}}{N} = \bar{y}_{..}, \quad \hat{\alpha}_j = \frac{Y_{j.}}{K} - \frac{Y_{..}}{N} = \bar{y}_{j.} - \bar{y}_{..}$$

and

$$\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = \frac{Y_{..}^2}{N} + \sum_{j=1}^J Y_{j.} \left(\frac{Y_{j.}}{K} - \frac{Y_{..}}{N} \right) = \frac{1}{K} \sum_{j=1}^J Y_{j.}^2$$

In this estimation, we have $\sum_{j=1}^J \hat{\alpha}_j = 0$ due to the constraint $\sum_{j=1}^J \alpha_j = 0$.

(ii) corner-point constraint

The corner point constraint assumes that any specific relative effect α_j is zero. For example, if we need one constraint, then we may assume $\alpha_1 = 0$. Under this constraint, we have

$$\mathbf{X}_{JK \times J} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{1} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \mu \\ \alpha_2 \\ \vdots \\ \alpha_J \end{bmatrix}$$

and therefore,

$$\mathbf{X}'\mathbf{X}_{J \times J} = \begin{bmatrix} N & K & K & \cdots & K \\ K & K & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ K & 0 & 0 & \cdots & K \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} Y_{..} \\ Y_{2.} \\ \vdots \\ Y_{J.} \end{bmatrix}$$

Now, $\mathbf{X}'\mathbf{X}$ becomes non-singular and the estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\mu} \\ \hat{\alpha}_2 \\ \vdots \\ \hat{\alpha}_J \end{bmatrix} = \frac{1}{K} \begin{bmatrix} Y_{1.} \\ Y_{2.} - Y_{1.} \\ \vdots \\ Y_{J.} - Y_{1.} \end{bmatrix}$$

Also, we have

$$\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = \frac{1}{K} \left[Y_{..} Y_{1.} + \sum_{j=2}^J Y_{j.} (Y_{j.} - Y_{1.}) \right] = \frac{1}{K} \sum_{j=1}^J Y_{j.}^2$$

Note that $D = (\mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y})/\sigma^2$, so that the deviance based on both methods (sum-to-zero constraint and the corner point constraint) are the same.

(3) Testing

To test whether there are treatment effects or not, consider

$$H_0 : \alpha_1 = \dots = \alpha_J = 0$$

then, under H_0 , we have $E(Y_{jk}) = \mu$, so that $\mathbf{X}'\mathbf{X} = N$, $\mathbf{X}'\mathbf{y} = Y_{..}$, $\hat{\beta} = \hat{\mu} = Y_{..} / N$. Hence,

$$D_0 = \frac{1}{\sigma^2}(\mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}) = \frac{1}{\sigma^2} \left(\sum_{j=1}^J \sum_{k=1}^K Y_{jk}^2 - \frac{Y_{..}^2}{N} \right)$$

which is $\chi^2(N - 1)$. On the other hand, under the current model,

$$D_1 = \frac{1}{\sigma^2}(\mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}) = \frac{1}{\sigma^2} \left(\sum_{j=1}^J \sum_{k=1}^K Y_{jk}^2 - \frac{1}{K} \sum_{j=1}^J Y_{j.}^2 \right)$$

which is $\chi^2(N - J)$. Therefore, the test statistic is

$$\Delta D = D_0 - D_1 = \frac{1}{\sigma^2} \left(\frac{1}{K} \sum_{j=1}^J Y_{j.}^2 - \frac{1}{N} Y_{..}^2 \right)$$

which is $\chi^2(J - 1)$. Hence, if σ^2 is known, then we reject H_0 if

$$\Delta D > \chi_{\alpha}^2(J - 1)$$

If σ^2 is unknown, we use F -statistic, i.e.,

$$F = \frac{D_0 - D_1}{(N - 1) - (N - J)} \bigg/ \frac{D_1}{(N - J)}$$

which follows $F(J - 1, N - J)$ distribution. Hence, we reject H_0 if $F >$

$F_{\alpha}(J - 1, N - J)$.

Ex.8.8 (p.321)

Heights of a plant for 3 types of fertilizers				
A	4.17	5.58	5.18	6.11
B	4.81	4.17	4.41	3.59
C	6.31	5.12	5.54	5.50

In this data, $J = 3$, $K = 4$, $N = 12$, and the ANOVA model is

$$E(Y_{jk}) = \mu + \alpha_j, \quad j = 1, 2, 3; k = 1, 2, 3, 4$$

and $\mathbf{X}$ and $\boldsymbol{\beta}$ are given by

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

and therefore,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 12 & 4 & 4 & 4 \\ 4 & 4 & 0 & 0 \\ 4 & 0 & 4 & 0 \\ 4 & 0 & 0 & 4 \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} Y_{..} \\ Y_{1.} \\ Y_{2.} \\ Y_{3.} \end{bmatrix} = \begin{bmatrix} 60.49 \\ 21.04 \\ 16.98 \\ 22.47 \end{bmatrix}$$

Since $\mathbf{X}'\mathbf{X}$ is singular, we assign the sum-to-zero constraint, $\alpha_1 + \alpha_2 + \alpha_3 =$

0, then

$$\hat{\beta} = \begin{bmatrix} \bar{Y}_{..} \\ \bar{Y}_{1.} - \bar{Y}_{..} \\ \bar{Y}_{2.} - \bar{Y}_{..} \\ \bar{Y}_{3.} - \bar{Y}_{..} \end{bmatrix} = \begin{bmatrix} 5.04 \\ 0.22 \\ -0.80 \\ 0.58 \end{bmatrix}$$

Also, we must have $\hat{\alpha}_1 + \hat{\alpha}_2 + \hat{\alpha}_3 = 0$. Now, $\hat{\beta}' X' y = 308.95$ gives $D = (y'y - \hat{\beta}' X' y) / \sigma^2 = (312.52 - 308.95) / \sigma^2 = 3.57 / \sigma^2$.

Next, under the corner-point constraint $\alpha_1 = 0$, X and β are given by

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} \mu \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

and therefore,

$$X'X = \begin{bmatrix} 12 & 4 & 4 \\ 4 & 4 & 0 \\ 4 & 0 & 4 \end{bmatrix}, \quad X'y = \begin{bmatrix} Y_{..} \\ Y_{2.} \\ Y_{3.} \end{bmatrix} = \begin{bmatrix} 60.49 \\ 16.98 \\ 22.47 \end{bmatrix}$$

Hence,

$$\hat{\beta} = \begin{bmatrix} \bar{Y}_{1.} \\ \bar{Y}_{2.} - \bar{Y}_{1.} \\ \bar{Y}_{3.} - \bar{Y}_{1.} \end{bmatrix} = \begin{bmatrix} 5.26 \\ -1.02 \\ 0.36 \end{bmatrix}$$

Note that $\hat{\beta}'X'y = 308.95$ which is the same as the value under sum-to-zero constraint.

Now, consider $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$, then the deviance under H_0 is

$$D_0 = \frac{1}{\sigma^2} \left(\sum_{j=1}^3 \sum_{k=1}^4 Y_{jk}^2 - \frac{Y_{..}^2}{12} \right) = 7.60 / \sigma^2$$

so that $\Delta D = D_0 - D_1 = 4.03 / \sigma^2$. Since σ^2 is unknown, we use F -statistic, i.e.,

$$F = \frac{(4.03 / \sigma^2)}{(11 - 9)} \bigg/ \frac{(3.57 / \sigma^2)}{9} = 5.08$$

which is larger than $F_{.05}(2, 9) = 4.26$, we reject H_0 . On the other hand, if we replace σ^2 in ΔD by its estimator $3.57/9 = 0.40$, then we have $\Delta D = 10.08$ which is larger than $\chi_{.05}^2(2) = 5.99$. Therefore, we have the same result as F -statistic.

ANOVA table of the heights of a plant for 3 fertilizers

s.v.	d.f.	SS	MS	F
treatment	2	4.03	2.02	5.08
error	9	3.57	0.40	
total	11	7.60		

Ex.8.9 (p.324)

Now, we extend to twoway classification defined as

$$Y_{jkl} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} + \varepsilon_{jkl}, \quad j = 1, \dots, J; \quad k = 1, \dots, K; \quad l = 1, \dots, L,$$

where μ is an overall effect, α_j is the main effect of factor A, β_k is the main effect of factor B, and $(\alpha\beta)_{jk}$ is the *interaction effect* of factors A and B. At each level, L replications are done. The data in Table 8.8 represents the effect of temperature and pressure on the viscosity of tire. Here, factor A is the temperature with $J = 3$, factor B is the pressure with $K = 2$, and $L = 2$ replications are done at each level. Therefore, the total number of observations is $N = JKL = 12$.

Table 8.8 Temperature, pressure, and viscosity of tire

temperature	pressure	B_1		B_2		total
A_1		6.8	6.6	5.3	6.1	24.8
A_2		7.5	7.4	7.2	6.5	28.6
A_3		7.8	9.1	8.8	9.1	34.8
합		45.2		43.0		88.2

Now, we are interested in 3 questions;

A : Is there the effect of temperature?

B : Is there the effect of pressure?

I : Is there the interaction effect of temperature and pressure?

which can be modelled as follows;

$$A : Y_{jkl} = \mu + \alpha_j + \varepsilon_{jkl}$$

$$B : Y_{jkl} = \mu + \beta_k + \varepsilon_{jkl}$$

$$I : Y_{jkl} = \mu + \alpha_j + \beta_k + \varepsilon_{jkl}$$

Now, under the full model, we have

$$X = \begin{bmatrix} 110010100000 \\ 110010100000 \\ 110001010000 \\ 110001010000 \\ 101010001000 \\ 101010001000 \\ 101001000100 \\ 101001000100 \\ 100110000010 \\ 100110000010 \\ 100101000001 \\ 100101000001 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \beta_1 \\ \beta_2 \\ (\alpha\beta)_{11} \\ (\alpha\beta)_{12} \\ (\alpha\beta)_{21} \\ (\alpha\beta)_{22} \\ (\alpha\beta)_{31} \\ (\alpha\beta)_{32} \end{bmatrix}, \quad X'y = \begin{bmatrix} Y_{...} \\ Y_{1..} \\ Y_{2..} \\ Y_{3..} \\ Y_{.1.} \\ Y_{.2.} \\ Y_{11.} \\ Y_{12.} \\ Y_{21.} \\ Y_{22.} \\ Y_{31.} \\ Y_{32.} \end{bmatrix} = \begin{bmatrix} 88.2 \\ 24.8 \\ 28.6 \\ 34.8 \\ 45.2 \\ 43.0 \\ 13.4 \\ 11.4 \\ 14.9 \\ 13.7 \\ 16.9 \\ 17.9 \end{bmatrix}$$

Note that X is 12×12 matrix, however, only 6 column vectors are linearly independent. Therefore, we need $6 = 12 - 6$ constraints. If we use the sum-to-zero constraints, then

$$\begin{aligned}\alpha_1 + \alpha_2 + \alpha_3 &= 0 & \beta_1 + \beta_2 &= 0 \\ (\alpha\beta)_{11} + (\alpha\beta)_{12} &= 0 & (\alpha\beta)_{21} + (\alpha\beta)_{22} &= 0 \\ (\alpha\beta)_{31} + (\alpha\beta)_{32} &= 0 & (\alpha\beta)_{11} + (\alpha\beta)_{21} + (\alpha\beta)_{31} &= 0\end{aligned}$$

Under this constraints, we have

$$\hat{\beta} = (7.35, -1.15, -0.2, 1.35, 0.18, -0.18, 0.32, -0.32, 0.12, -0.12, -0.43, 0.43)'$$

and $\hat{\beta}'X'y = 662.62$.

On the other hand, the corner point constraints can be written as

$$\alpha_1 = \beta_1 = (\alpha\beta)_{11} = (\alpha\beta)_{12} = (\alpha\beta)_{21} = (\alpha\beta)_{31} = 0$$

and under this constraints, we have

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} \mu \\ \alpha_2 \\ \alpha_3 \\ \beta_2 \\ (\alpha\beta)_{22} \\ (\alpha\beta)_{32} \end{bmatrix}$$

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} Y_{...} \\ Y_{2..} \\ Y_{3..} \\ Y_{12.} \\ Y_{22.} \\ Y_{32.} \end{bmatrix} = \begin{bmatrix} 88.2 \\ 28.6 \\ 34.8 \\ 43.0 \\ 13.7 \\ 17.9 \end{bmatrix}$$

Therefore, we have $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (6.7, 0.75, 1.75, -1.0, 0.4, 1.5)'$ and $\hat{\beta}'\mathbf{X}'\mathbf{y} = 662.62$ which is the same as the value obtained under the sum-to-zero constraints.

Finally, we can test 3 current models A , B , I , and these can be done by the following ANOVA table.

Table 8.9 ANOVA table for the viscosity of tire data

s.v.	d.f.	S.S.	MS	F
$A(\text{temperature})$	2	12.74	6.37	25.82
$B(\text{pressure})$	1	0.40	0.40	1.63
$A \times B$	2	1.21	0.60	2.45
error	6	1.48	0.25	
total	11	15.83		

We see that the main effect of temperature is significant, however, the main effect of pressure and the interaction effect of temperature and pressure are not significant.

Chapter 9. Generalized Linear Models II

9.1 Logistic Regression Model

9.1.1 Binomial distribution and binary data

$Z \sim B(1, \pi_i)$: binary data or dichotomous data

$(x_{i1}, \dots, x_{ip})'$: i th covariate

$Y_i \sim B(m_i, \pi_i)$: i th response

Table 9.1 Insecticide data

amount of insecticide(x_i)	no. of bugs(m_i)	no. of dead bugs(y_i)
1.6907	59	6
1.7242	60	13
1.7552	62	18
1.7842	56	28
1.8113	63	52
1.8369	59	53
1.8610	62	61
1.8839	60	60

The simple linear regression model

$$\pi_i = \beta_0 + \beta_1 X_i, \quad i = 1, \dots, n$$

is not appropriate because either $\hat{\pi}_i < 0$ or $\hat{\pi}_i > 1$.

9.1.2 Logistic regression model

Consider a link function $g(\pi_i) = x_i'\beta$ such that $0 < \hat{\pi}_i < 1$ is guaranteed.

Note that $\pi_i = g^{-1}(x_i'\beta)$. Hence, if we take g^{-1} as a cdf, then, $0 < \hat{\pi}_i < 1$ is guaranteed. i.e.,

$$\pi = g^{-1}(x'\beta) = \int_{-\infty}^{\eta} f(s)ds,$$

where $f(s)$ is a pdf.

If $f(s)$ is a standard normal pdf and Φ is the normal cdf, then

$$\pi = \Phi(x'\beta)$$

i.e.,

$$\Phi^{-1}(\pi) = x'\beta$$

which is called probit model.

On the other hand, if $f(s)$ is a logistic pdf, i.e., $f(s) = \frac{e^s}{(1 + e^s)^2}$, then

$$\pi = \int_{-\infty}^{\eta} f(s)ds = \frac{e^{\eta}}{1 + e^{\eta}}$$

Hence, if we plug in $\eta = x'\beta$, then

$$\pi = \frac{\exp(x'\beta)}{1 + \exp(x'\beta)}$$

so that

$$\log\left(\frac{\pi}{1 - \pi}\right) = x'\beta$$

Because

$$\text{logit } \pi = \log \left(\frac{\pi}{1 - \pi} \right)$$

which is called logit link, and this model is called logistic regression model.

Here, we investigate the logistic regression model in detail. Let $Y_1, \dots, Y_n$ be independent $B(m_i, \pi_i)$, then the log-likelihood is

$$l = \sum_{i=1}^n \left\{ y_i \log \left(\frac{\pi_i}{1 - \pi_i} \right) + m_i \log(1 - \pi_i) + \log \binom{m_i}{y_i} \right\}$$

For the one covariate case,

$$\log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 x_i$$

so that

$$l = \sum_{i=1}^n \left\{ y_i(\beta_0 + \beta_1 x_i) - m_i \log[1 + \exp(\beta_0 + \beta_1 x_i)] + \log \binom{m_i}{y_i} \right\}.$$

Now, the score function is

$$\begin{aligned} U_1 &= \frac{\partial l}{\partial \beta_0} = \sum \left\{ y_i - m_i \left[\frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)} \right] \right\} = \sum (y_i - m_i \pi_i) \\ U_2 &= \frac{\partial l}{\partial \beta_1} = \sum \left\{ y_i x_i - m_i x_i \left[\frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)} \right] \right\} = \sum x_i (y_i - m_i \pi_i) \end{aligned}$$

and the Fisher information is

$$I = \begin{pmatrix} \sum m_i \pi_i (1 - \pi_i) & \sum m_i x_i \pi_i (1 - \pi_i) \\ \sum m_i x_i \pi_i (1 - \pi_i) & \sum m_i x_i^2 \pi_i (1 - \pi_i) \end{pmatrix}$$

Hence, by the Newton-Raphson method, we have

$$\hat{\beta}^{(t+1)} = \hat{\beta}^{(t)} + [\mathbf{I}^{(t)}]^{-1} \mathbf{U}^{(t)}, \quad t = 0, 1, 2, \dots$$

Ex.9.1(p.334), Ex.9.2(p.335)

Table 9.4 Seed germination data

Storage type	Centrifugal force (g)			
		40	150	350
Control	y_{1k}	55	52	57
	n_{1k}	102	99	108
Treatment	y_{2k}	55	50	50
	n_{2k}	76	81	90

Consider three types of models;

$$\text{Model 1 : } \text{logit } \pi_{jk} = \alpha_j + \beta_j x_k$$

$$\text{Model 2 : } \text{logit } \pi_{jk} = \alpha_j + \beta x_k$$

$$\text{Model 3 : } \text{logit } \pi_{jk} = \alpha + \beta x_k$$

Table 9.5. Three types of logistic models for the seed germination data. Coefficients fit (s.e.) and deviance.

Model 1	Model 2	Model 3
$a_1 = .238(.628)$	$a_1 = .877(.487)$	$a = 1.021(.481)$
$a_2 - a_1 = 1.977(.998)$	$a_2 - a_1 = .407(.175)$	$b = -.148(.096)$
$b_1 = -.023(.127)$	$b = -.155(.097)$	
$b_2 - b_1 = -.319(.199)$		
$D_1 = .0277$	$D_2 = 2.619$	$D_3 = 8.092$

If we compare Model 3 with Model 2, then $\Delta D = D_3 - D_2 = 5.473$ which is larger than $\chi^2_{0.05}(1) = 3.84$, so that Model 2 is more appropriate than Model 3. On the other hand, if we compare Model 2 with Model 1, then $\Delta D = D_2 - D_1 = 2.591$ which is not larger than $\chi^2_{0.05}(1)$, so that Model 2 is not worse than the Model 1. Conclusively, we recommend choosing Model 2.

9.1.3 Estimation of Median

In the univariate logistic regression model, we are often interested in estimating the covariate x for $\hat{\pi} = 0.5$. This value is called *median effective dose* or *median lethal dose* in medical statistics. Let

$$\log(\pi/(1 - \pi)) = \beta_0 + \beta_1 x$$

then the point estimate of median satisfying $\hat{\pi} = 0.5$ is

$$\hat{\theta}_{.5} = -\hat{\beta}_0/\hat{\beta}_1.$$

To obtain the C.I. for the median, we need to compute $Var(\hat{\theta}_{.5})$. To do this, we use the multivariate delta method, i.e.,

$$Var[g(\hat{\beta}_0, \hat{\beta}_1)] \simeq \left(\frac{\partial g}{\partial \hat{\beta}_0}\right)^2 Var(\hat{\beta}_0) + \left(\frac{\partial g}{\partial \hat{\beta}_1}\right)^2 Var(\hat{\beta}_1) + 2\left(\frac{\partial^2 g}{\partial \hat{\beta}_0 \partial \hat{\beta}_1}\right) Cov(\hat{\beta}_0, \hat{\beta}_1)$$

so that

$$Var(\hat{\theta}_{.5}) \simeq (v_{00} - 2\hat{\rho}v_{01} + \hat{\rho}^2v_{11})/\hat{\beta}_1^2,$$

where

$$v_{00} = Var(\hat{\beta}_0), \hat{\rho} = \hat{\beta}_0/\hat{\beta}_1, v_{01} = Cov(\hat{\beta}_0, \hat{\beta}_1), v_{11} = Var(\hat{\beta}_1).$$

Therefore, the $100(1 - \alpha)\%$ C.I. for $\theta_{.5}$ is

$$\hat{\theta}_{.5} \pm z_{\alpha/2} s.e.(\hat{\theta}_{.5}),$$

where $s.e.(\hat{\theta}_{.5}) = \{Var(\hat{\theta}_{.5})\}^{1/2}$.

9.1.4 The use of LSE

We introduce the use of least squares in estimating β instead of the Newton-Raphson method. Let $Y_i \sim B(m_i, \pi_i)$, $i = 1, \dots, n$ and $p_i = y_i / m_i$ be the sample proportion. Now, by the Taylor expansion of $\psi(p_i)$ w.r.t. π_i ,

$$\psi(p_i) = \psi(\pi_i) + (p_i - \pi_i)\psi'(\pi_i) + o(m_i^{-2})$$

Now, since $E(p_i) = \pi_i$, we have $E(\psi(p_i)) = \psi(\pi_i)$, and

$$\begin{aligned} \text{Var}(\psi(p_i)) &\simeq \text{Var}[(p_i - \pi_i)\psi'(\pi_i)] \\ &= \psi'(\pi_i)^2 \frac{\pi_i(1 - \pi_i)}{m_i} \end{aligned}$$

Now, the Pearson X^2 is

$$X^2 = \sum_{i=1}^n \frac{\{\psi(p_i) - \psi(\pi_i)\}^2}{\psi'(\pi_i)^2 \pi_i(1 - \pi_i) / m_i}$$

Table 9.7 Types of LSE

$\psi(\pi_i)$	X^2
π_i	$\sum \frac{(p_i - \pi_i)^2}{\pi_i(1 - \pi_i) / m_i}$
$\text{logit } \pi_i$	$\sum \{(\text{logit } p_i - \text{logit } \pi_i)^2 \pi_i(1 - \pi_i) m_i\}$
$\arcsin \sqrt{\pi_i}$	$\sum 4m_i \{\arcsin \sqrt{p_i} - \arcsin \sqrt{\pi_i}\}^2$

First, for the case of $\psi(\pi_i) = \pi_i$, we have $\pi_i = \mathbf{x}_i' \boldsymbol{\beta}$, so that

$$X^2 = \sum \frac{(p_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{\pi_i(1 - \pi_i) / m_i}$$

and if we replace π_i by p_i , then we have a modified X_{mod}^2 , i.e.,

$$X_{mod}^2 = \sum \frac{(p_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{p_i(1 - p_i) / m_i}$$

Now, the fitted value is $\hat{\pi}_i = \mathbf{x}_i' \hat{\boldsymbol{\beta}}$, however, it is not desirable because $\hat{\pi}_i$ could be outside the interval $(0, 1)$.

Second, consider $\psi(\pi_i) = \text{logit } \pi_i$, i.e., $\pi_i = \exp(\mathbf{x}_i' \boldsymbol{\beta}) / [1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})]$, then

$$X_{mod}^2 = \sum (\text{logit } p_i - \mathbf{x}_i' \boldsymbol{\beta})^2 p_i(1 - p_i) m_i$$

Cox (1970) suggested the *empirical logistic transformation*

$$z_i = \log \left(\frac{y_i + \frac{1}{2}}{m_i - y_i + \frac{1}{2}} \right)$$

instead of $\text{logit } p_i$. In this case, the fitted value $\hat{\pi}_i = \exp(\mathbf{x}_i' \hat{\boldsymbol{\beta}}) / [1 + \exp(\mathbf{x}_i' \hat{\boldsymbol{\beta}})]$ is always between 0 and 1.

Finally, the arcsine transformation comes from the variance stabilizing transformation, i.e.,

$$\text{Var}[\psi(p_i)] \simeq \psi'(\pi_i)^2 \pi_i(1 - \pi_i) / m_i \equiv \text{constant}$$

and we have

$$\psi(\pi_i) = \arcsin \sqrt{\pi_i}$$

and

$$\text{Var}[\arcsin \sqrt{p_i}] = 1/4m_i.$$

9.1.5 Multinomial distribution and polytomous data

Assume that the response Y has k possible categories, i.e, the response is polytomous data. Let $\mathbf{x}_i = (X_{i1}, \dots, X_{ip})'$ and π_j be the probability that Y_i is in the j th category, so that $\sum_{j=1}^k \pi_j = 1$. Also, let $\gamma_j = \pi_1 + \dots + \pi_j$ be the j th cumulative probability. Then, the logistic regression can be modelled as

$$\log \left(\frac{\gamma_j(\mathbf{x})}{1 - \gamma_j(\mathbf{x})} \right) = \theta_j - \boldsymbol{\beta}'\mathbf{x}, \quad j = 1, \dots, k-1,$$

where $\gamma_j(\mathbf{x}) = P(Y \leq j \mid \mathbf{x})$ and θ_j is the j th intercept term. This model is often called the *proportional – odds model* because the odds ratio at $\mathbf{x} = \mathbf{x}_1$ and $\mathbf{x} = \mathbf{x}_2$ is given by

$$\frac{\gamma_j(\mathbf{x}_1) / (1 - \gamma_j(\mathbf{x}_1))}{\gamma_j(\mathbf{x}_2) / (1 - \gamma_j(\mathbf{x}_2))} = \exp\{-\boldsymbol{\beta}'(\mathbf{x}_1 - \mathbf{x}_2)\}$$

which does not depend on a specific category j .

Ex.9.3 (p.340)

Table 9.8. Taste data for 4 types of cheese

favor cheese	1	2	3	4	5	6	7	8	9	total
A	0	0	1	7	8	8	19	8	1	52
B	6	9	12	11	7	6	1	0	0	52
C	1	1	6	8	23	7	5	1	0	52
D	0	0	0	1	3	7	14	16	11	52
total	7	10	19	27	41	28	39	25	12	208

The proportional odds model becomes

$$\text{logit } \gamma_{ij} = \theta_j - \beta_i, \quad i = 1, 2, 3, 4; \quad j = 1, \dots, 8$$

and the fitted values using the corner point constraint ($\beta_4 = 0$) are $\hat{\beta}_1 = -1.613(0.378)$, $\hat{\beta}_2 = -4.965(0.474)$, $\hat{\beta}_3 = -3.323(0.425)$, $\hat{\beta}_4 = 0$. (values in the parenthesis denote the standard error). Therefore, the order of favor for 4 types of cheese is D, A, C, B.

Under the null hypothesis ($\beta = 0$), the deviance is 168.8, and the d.f. is $(9 - 1) \times 4 - 8 = 24$, and under the current model the deviance is 20.31 and d.f. is $24 - 3 = 21$, so that we reject H_0 , i.e., there is favor difference for each cheese.

Ex.9.4(p.341)

pneumoconiosis data

response : 3 category (normal, moderate, severe)

covariate : no. of years served in coal mine

Table 9.9 No. of years served in coal mine and 3 category (normal, moderate, severe)

years served	normal	moderate	severe
5.8	98	0	0
15.0	51	2	1
21.5	34	6	3
27.5	35	5	8
33.5	32	10	9
39.5	23	7	8
46.0	12	6	1
51.5	4	2	5

Model :

$$\text{logit } \gamma_{ij} = \theta_j - \beta \log x_i, \quad j = 1, 2; \quad i = 1, \dots, 8$$

estimators : $\hat{\beta} = 2.60(.38), \hat{\theta}_1 = 9.68, \hat{\theta}_2 = 10.58$

For example, the risk of moderate pneumoconiosis for 5 years served coal miner is 1 out of $\exp(9.68 - 2.60 \log 5) = 243$ miners. For 10 years, it will be 1 out of 40 miners, and for 20 years, 1 out of 7. The risk of severe pneumoconiosis for 5 years served coal miner is 1 out of $\exp(10.58 - 2.60 \log 5) = 600$ miners, and for 20 years, it will be 1 out of 17.

9.2 Log-Linear Model

9.2.1 Categorical data

Ex.9.5 (p.342)

malignant tumor data for $n = 400$ patients

two factors : site and type of tumor data

Table 9.10. Frequency table for the malignant tumor

type site	face	body	arm, leg	total
Hutchinson tumor	22	2	10	34
epidermal tumor	16	54	115	185
nodular tumor	19	33	73	125
others	11	17	28	56
total	68	106	226	400

Ex.9.6(p.342)

Influenza vaccine data

treatment group : inject vaccine to 35 people

control group : inject placebo to 38 people

Compare HIA(Haemagglutinin Inhibiting Antibody) 6 weeks after injection :

prospective study

Table 9.11 Influenza vaccine data

HIA	small	moderate	large	total
control	25	8	5	38
treatment	6	18	11	35
total	31	26	16	73

Ex.9.7 (p.343)

Side effect of aspirin on the stomach and duodenal ulcer

Investigate the use of aspirin for the 64 case (stomach ulcer patients) group and the 68 control (normal group). Also, investigate the use of aspirin for the 57 case (duodenal ulcer patients) group and the 61 control (normal group). We want to see whether there exists the side effect of aspirin use on ulcer. It is a typical example for the *reterspective study*.

Table 9.12. Case-control study for the side effect of aspirin on the stomach and duodenal ulcer

	aspirin		total
	did not take	took	
stomach case	39	25	64
control	62	6	68
duodenal case	49	8	57
control	53	8	61

9.2.2 Poisson distribution and multinomial distribution

Consider the following $J \times K$ contingency table

Table 9.13. $J \times K$ contingency table

	B_1	B_2	$\cdots$	B_K	total
A_1	Y_{11}	Y_{12}	$\cdots$	Y_{1K}	$Y_{1\cdot}$
A_2	Y_{21}	Y_{22}	$\cdots$	Y_{2K}	$Y_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
A_J	Y_{J1}	Y_{J2}	$\cdots$	Y_{JK}	$Y_{J\cdot}$
total	$Y_{\cdot 1}$	$Y_{\cdot 2}$	$\cdots$	$Y_{\cdot K}$	$n = Y_{\cdot\cdot}$

(1) No margin fixed case

Assume that Y'_{jk} s are independent and $Y_{jk} \sim P(\lambda_{jk})$, then the jpdf becomes

$$f(\mathbf{y}; \boldsymbol{\lambda}) = \prod_{j=1}^J \prod_{k=1}^K \frac{\lambda_{jk}^{y_{jk}} e^{-\lambda_{jk}}}{y_{jk}!}$$

(2) Fixed total case

First, consider a very simple case. Let Y_1 and Y_2 be independent Poisson with mean λ_1 and λ_2 , respectively. Then, $Y_1 + Y_2$ is Poisson with mean $\lambda_1 + \lambda_2$. Now, it is easy to show that the conditional distribution of Y_1 given $Y_1 + Y_2 = n$ is $B(n, \lambda_1 / (\lambda_1 + \lambda_2))$. Now, we can extend this result, i.e.,

$$\begin{aligned} f(\mathbf{y}; n) &= \prod_{j=1}^J \prod_{k=1}^K \frac{\lambda_{jk}^{y_{jk}} e^{-\lambda_{jk}}}{y_{jk}!} / \frac{\lambda_{..}^n e^{-\lambda_{..}}}{n!} \\ &= n! \prod_{j=1}^J \prod_{k=1}^K \frac{\theta_{jk}^{y_{jk}}}{y_{jk}!} \end{aligned}$$

where $\lambda_{..} = \sum \sum \lambda_{jk}$, $\theta_{jk} = \lambda_{jk} / \lambda_{..}$. Therefore, it is multinomial distribution.

(3) One margin fixed case

The conditional distribution of $Y_{j1}, \dots, Y_{jK}$ given Y_j is

$$\begin{aligned} f(y_{j1}, \dots, y_{jK} | \sum_{j=1}^J y_{j\cdot}) &= \prod_{k=1}^K \frac{\lambda_{jk}^{y_{jk}} e^{-\lambda_{jk}}}{y_{jk}!} / \frac{\lambda_{j\cdot}^{y_{j\cdot}} e^{-\lambda_{j\cdot}}}{y_{j\cdot}!} \\ &= y_{j\cdot}! \prod_{k=1}^K \theta_{jk}^{y_{jk}} / y_{jk}! \end{aligned}$$

where $\theta_{jk} = \lambda_{jk} / \lambda_{j\cdot}$, $\sum_k \theta_{jk} = 1$, and it can be rewritten as

$$f(\mathbf{y} | y_{j\cdot}, j = 1, \dots, J) = \prod_{j=1}^J y_{j\cdot}! \prod_{k=1}^K \theta_{jk}^{y_{jk}} / y_{jk}!$$

which is called *product multinomial distribution*.

(4) Extension to 3-dim contingency table

We can extend the above results to the $J \times K \times L$ 3-dimensional contingency table. Let

$$Y_{jkl} (j = 1, \dots, J; k = 1, \dots, K; l = 1, \dots, L)$$

and assume that Y_{jkl} has Poisson distribution with mean λ_{jkl} , then

we have the following results.

$$f(\mathbf{y} : \boldsymbol{\lambda}) = \prod_{j=1}^J \prod_{k=1}^K \prod_{l=1}^L \lambda_{jkl}^{y_{jkl}} e^{-\lambda_{jkl}} / y_{jkl}!$$

$$\begin{aligned}
f(\mathbf{y} \mid n) &= \prod_{j=1}^J \prod_{k=1}^K \prod_{l=1}^L \frac{\lambda_{jkl}^{y_{jkl}} e^{-\lambda_{jkl}}}{y_{jkl}!} / \frac{\lambda_{...}^n e^{-\lambda_{...}}}{n!} \\
&= n! \prod_{j=1}^J \prod_{k=1}^K \prod_{l=1}^L \frac{\theta_{jkl}^{y_{jkl}}}{y_{jkl}!}
\end{aligned}$$

where $\theta_{jkl} = \lambda_{jkl} / \lambda_{...}$, $\sum \sum \sum \theta_{jkl} = 1$. Also,

$$f(\mathbf{y} \mid y_{..l}, l = 1, \dots, L) = \prod_{l=1}^L y_{..l}! \prod_{j=1}^J \prod_{k=1}^K \theta_{jkl}^{y_{jkl}} / y_{jkl}!$$

$$f(\mathbf{y} \mid y_{j\cdot l}, j = 1, \dots, J; l = 1, \dots, L) = \prod_{j=1}^J \prod_{l=1}^L y_{j\cdot l}! \prod_{k=1}^K \theta_{jkl}^{y_{jkl}} / y_{jkl}!$$

For example, in the aspirin data, we have $J = 2$ (case, control), $K = 2$ (use of aspirin or not), $L = 2$ (stomach, duodenal), and two margins (J and L) are fixed.

9.2.3 Log-linear model

If Y_{jk} follows a Poisson distribution, then the natural link function is log function. Let $\mu = E(Y)$, then

$$\eta = \log \mu = \mathbf{x}' \boldsymbol{\beta}$$

so that, in $J \times K$ contingency table, if two factors are independent, then

$$\eta_{jk} = \log E(Y_{jk}) = \mu + \alpha_j + \beta_k$$

and, under the maximal model,

$$\eta_{jk} = \log E(Y_{jk}) = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk}$$

which are called *log – linear model*

Under the maximal model, $\hat{\mu}_{jk} = y_{jk}$, hence the log-likelihood becomes

$$l(\hat{\beta}_{max} : y) = \text{constant} + \sum \sum y_{jk} \log y_{jk} - \sum \sum y_{jk}$$

and under the current model,

$$l(\hat{\beta} : y) = \text{constant} + \sum \sum y_{jk} \log \hat{\mu}_{jk} - \sum \sum \hat{\mu}_{jk},$$

where $\hat{\mu}_{jk}$ is estimator of $\mu_{jk} = E(Y_{jk})$. Therefore, the deviance becomes

$$D = 2 \sum \sum y_{jk} \log(y_{jk} / \hat{\mu}_{jk}) - 2 \sum \sum (y_{jk} - \hat{\mu}_{jk})$$

On the other hand, the Pearson's chi-square is

$$X^2 = \sum \sum \frac{(y_{jk} - \hat{\mu}_{jk})^2}{\hat{\mu}_{jk}}$$

Note that both D and X^2 follow asymptotically chi-square distribution.

Ex. 9.5 (p. 348)

Consider the malignant tumor data. By the log-linear model, we have

$$\eta_{jk} = \log E(Y_{jk}) = \mu + \alpha_j + \beta_k, \quad j = 1, \dots, 4; k = 1, 2, 3$$

But, the inverse of $X'X$ does not exist, so that we use either the sum-to-zero-constraint ($\sum \alpha_j = 0, \sum \beta_k = 0$) or the corner point restriction ($\alpha_1 = 0, \beta_1 = 0$). Hence, under these restrictions, the number of parameters to be estimated is $1 + (J - 1) + (K - 1) = J + K - 1 = 6$. The maximal model is

$$\eta_{jk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk}$$

and the number of parameters to be estimated is $1 + (J - 1) + (K - 1) + (J - 1)(K - 1) = 12$ which is the same as the number of responses y_{jk} , i.e., $\hat{\eta}_{jk} = \log \hat{\mu}_{jk} = \log y_{jk}$. Therefore, $D = 0$.

Table 9.14 (p. 349)

Corner point restriction was used. Hence, $\text{Type}(1) = \text{Site}(1) = 0$. Also, note that $\hat{\mu} = 1.754$, $SE(\hat{\mu}) = 0.2036$, $\hat{\alpha}_2 = 1.694$, $SE(\hat{\alpha}_2) = 0.1862$.

Let's check for the fitted value of $y_{11} = 22$. Note that

$$\hat{\mu}_{11} = \hat{\mu} + \hat{\alpha}_1 + \hat{\beta}_1 = 1.754 + 0 + 0 = 1.754$$

$$\hat{y}_{11} = \exp(\hat{\eta}_{11}) = e^{1.754} = 5.78$$

Next, let's check the residuals. For example, the Pearson type residual of the 1st observation is

$$r_{11} = \frac{y_{11} - \hat{y}_{11}}{\sqrt{\hat{y}_{11}}} = \frac{22 - 5.78}{\sqrt{5.78}} = 6.747$$

Finally, $D = 51.795$, $X^2 = 65.813$ which are larger than $\chi^2_{0.05}(n - p) = \chi^2_{0.05}(6) = 12.59$, we reject H_0 , i.e, the occurrences of each tumor type is different at each site.

Ex.9.8(p.350)

Table 9.15 Fit of the log-linear model for he stomach and duodenal ulcer data

model	deviance	<i>d.f.</i>
$\mu + \alpha + \beta + \gamma$	21.79	4
$\mu + \alpha + \beta + \gamma + \alpha\beta$	10.54	3
$\mu + \alpha + \beta + \gamma + \alpha\gamma$	21.79	3
$\mu + \alpha + \beta + \gamma + \beta\gamma$	21.70	3
$\mu + \alpha + \beta + \gamma + \alpha\beta + \alpha\gamma$	10.54	2
$\mu + \alpha + \beta + \gamma + \alpha\beta + \beta\gamma$	6.45	2
$\mu + \alpha + \beta + \gamma + \alpha\gamma + \beta\gamma$	17.70	2
$\mu + \alpha + \beta + \gamma + \alpha\beta + \alpha\gamma + \beta\gamma$	6.28	1

9.3 Overdispersion

9.3.1 Binomial distribution and overdispersion

Note that population is often consists of many clusters, for example, family, households, and litter. *Overdispersion* occurs quite often in clusters.

Assume that there are k clusters and we have m observations from each cluster. Let X_i be the number of successes out of m observations, i.e., $X_i \sim B(m, \pi_i), i = 1, \dots, k$. Also, let $Y = \sum_{i=1}^k X_i$ be the total number of successes in the all clusters. In this case, the probability of success π_i could be different at each cluster, so that we regard π_i as a r.v. instead of a constant. Hence, we assume that

$$X_i \mid \pi_i \sim B(m_i, \pi_i)$$

and

$$E(\pi_i) = \pi, \text{ Var}(\pi) = \sigma^2 \pi(1 - \pi)$$

Let $n = mk$, then we have

$$E(Y) = n\pi$$

$$\text{Var}(Y) = n\pi(1 - \pi)\{1 + (m - 1)\sigma^2\}$$

which are proved in the followings; First, note that

$$\begin{aligned} E(Y) &= \sum E(X_i) \\ &= \sum E[E(X_i \mid \pi_i)] \\ &= \sum E[m\pi_i] \\ &= mk\pi \end{aligned}$$

and

$$\text{Var}(Y) = \sum \text{Var}(X_i)$$

Now, to compute $\text{Var}(X_i)$, we use the following equality. For two r.v.'s X and Z , we have

$$\text{Var}(X) = E[\text{Var}(X \mid Z)] + \text{Var}[E(X \mid Z)]$$

because

$$\begin{aligned} \text{Var}(X) &= E[(X - E(X))^2] \\ &= E[E\{(X - E(X))^2 \mid Z\}] \\ &= E[E\{(X - E(X \mid Z) + E(X \mid Z) - E(X))^2 \mid Z\}] \\ &= E[E\{(X - E(X \mid Z))^2 \mid Z\} + E\{(E(X \mid Z) - E(X))^2 \mid z\}] \\ &= E[\text{Var}(X \mid Z)] + \text{Var}[E(X \mid Z)] \end{aligned}$$

Therefore,

$$\begin{aligned}
Var(X_i) &= E[Var(X_i | \pi_i)] + Var[E(X_i | \pi_i)] \\
&= E[m\pi_i(1 - \pi_i)] + Var[m\pi_i] \\
&= mE[\pi_i - \pi_i^2] + m^2Var[\pi_i] \\
&= m\pi - m(\pi^2 + \sigma^2\pi(1 - \pi)) + m^2\sigma^2\pi(1 - \pi) \\
&= m\pi(1 - \pi)\{1 + (m - 1)\sigma^2\}
\end{aligned}$$

Here, we call $\phi = 1 + (m - 1)\sigma^2$ *dispersion parameter*. We say that if ϕ is larger than 1, then overdispersion occurred.

The other measure for detecting the overdispersion is done by Lexis (1879), and he suggested *Lexis ratio* defined as

$$Q = \frac{\sum_{i=1}^k (\hat{\pi}_i - \hat{\pi})^2 / k}{\hat{\pi}(1 - \hat{\pi}) / n}$$

where $\hat{\pi}_i = X_i / m$, $\hat{\pi} = Y / n$. We say the overdispersion occurred if $Q > 1$. The numerator is sample variance and the denominator is nominal variance, so that the overdispersion occurs when the sample variance is larger than the nominal variance.

Ex9.9(p.353)

[Table 9.17. The number of male children in 6115 households with 12 children](#)

no. of male children	0	1	2	3	4	5	6
no. of households	3	24	104	286	670	1033	1343
no. of male children	7	8	9	10	11	12	
no. of households	1112	829	478	181	45	7	

Here, $k = 6115$, $m = 12$, and the Lexis ratio is $Q = 1.165$, and therefore, there is some overdispersion in this data.

9.3.2 Poisson distribution and overdispersion

The overdispersion can also occur in the Poisson distribution. If $\text{Var}(Y) > E(Y)$ in Poisson, then the overdispersion might occur. Let Z_i , $i = 1, 2, \dots, N$ be the number of occurrences at the i th cluster, and the number of clusters N is also a r.v. which has Poisson distribution. In specific, let Y be the total number of occurrences in N clusters, i.e.,

$$Y = Z_1 + \dots + Z_N$$

We assume that Z and N are independent because they are Poisson process. Hence,

$$E(Y) = E[E(Y | N)] = E[NE(Z)] = E(N)E(Z)$$

and

$$\begin{aligned}
 \text{Var}(Y) &= E[\text{Var}(Y \mid N)] + \text{Var}[E(Y \mid N)] \\
 &= E[N\text{Var}(Z)] + \text{Var}[NE(Z)] \\
 &= E(N)\text{Var}(Z) + \text{Var}(N)\{E(Z)\}^2 \\
 &= E(N)E(Z^2)
 \end{aligned}$$

so that if $E(Z^2) > E(Z)$, then the overdispersion occurs.

Ex.9.10(p.354)

Table 9.18. Number of car accidents per year in 9461 Belgian drivers

no. of accidents	0	1	2	3	4	5	6	7
no. of drivers	7840	1317	239	42	14	4	4	1

sample variance = 0.289 ; nominal variance = 0.214 => overdispersion exists

9.4 Log-Linear Model and Logistic Regression Model

In $J \times K$ contingency table, the response Y_{jk} can be modelled by the log-linear model

$$\log \mu_{jk} = \varphi_j + \mathbf{x}'_{jk} \boldsymbol{\beta}$$

where $\mu_{jk} = E(Y_{jk})$ and $\mathbf{x}_{jk}$ is p -dimensional covariate, $\boldsymbol{\beta}$ is parameter of interest, and $\varphi_1, \dots, \varphi_J$ are incidental parameters. Hence, the log-likelihood is

$$\begin{aligned} l_Y(\boldsymbol{\varphi}, \boldsymbol{\beta}) &= \sum_j \sum_k \{y_{jk}(\varphi_j + \mathbf{x}'_{jk} \boldsymbol{\beta}) - \exp(\varphi_j + \mathbf{x}'_{jk} \boldsymbol{\beta})\} \\ &= \sum_j \varphi_j y_{j\cdot} + \sum_j \sum_k y_{jk} \mathbf{x}'_{jk} \boldsymbol{\beta} - \sum_j \sum_k \exp(\varphi_j + \mathbf{x}'_{jk} \boldsymbol{\beta}) \end{aligned}$$

Now, let $m_j = y_{j\cdot}$ and $\tau_j = \sum_k \mu_{jk} = \sum_k \exp(\varphi_j + \mathbf{x}'_{jk} \boldsymbol{\beta})$, then the log-likelihood can be expressed as a function of $(\boldsymbol{\tau}, \boldsymbol{\beta})$ instead of $(\boldsymbol{\varphi}, \boldsymbol{\beta})$, i.e.,

$$\begin{aligned} l_Y(\boldsymbol{\tau}, \boldsymbol{\beta}) &= \sum_j (m_j \log \tau_j - \tau_j) + \sum_j \{ \sum_k y_{jk} \mathbf{x}'_{jk} \boldsymbol{\beta} - m_j \log(\sum_k \exp(\mathbf{x}'_{jk} \boldsymbol{\beta})) \} \\ &= l_{\mathbf{m}}(\boldsymbol{\tau}; \mathbf{m}) + l_{Y|\mathbf{m}}(\boldsymbol{\beta}; \mathbf{y}) \end{aligned}$$

where $l_{\mathbf{m}}(\boldsymbol{\tau}; \mathbf{m})$ is due to $Y_{j\cdot} \sim P(\tau_j)$, and $l_{Y|\mathbf{m}}(\boldsymbol{\beta}; \mathbf{y})$ is the conditional log-likelihood when $\mathbf{m} = (Y_{1\cdot}, \dots, Y_{J\cdot})'$ is given. Now, all the information on $\boldsymbol{\beta}$ is contained in $l_{Y|\mathbf{m}}(\boldsymbol{\beta}; \mathbf{y})$, so that we maximize $l_{Y|\mathbf{m}}(\boldsymbol{\beta}; \mathbf{y})$ to get $\boldsymbol{\beta}$.

Let $Y_{jk} \sim M(Y_{j\cdot} ; \pi_{j1}, \dots, \pi_{jk})$, where

$$\pi_{jk} = \exp(\mathbf{x}'_{jk}\boldsymbol{\beta}) / \sum_k \exp(\mathbf{x}'_{jk}\boldsymbol{\beta})$$

then the log-likelihood for $\mathbf{y}$ is

$$\begin{aligned} \sum_j \sum_k y_{jk} \log \pi_{jk} &= \sum_j \left\{ \sum_k y_{jk} \mathbf{x}'_{jk} \boldsymbol{\beta} - \sum_k y_{jk} \log \sum_k \exp(\mathbf{x}'_{jk} \boldsymbol{\beta}) \right\} \\ &= \sum_j \left\{ \sum_k y_{jk} \mathbf{x}'_{jk} \boldsymbol{\beta} - m_j \log \left(\sum_k \exp(\mathbf{x}'_{jk} \boldsymbol{\beta}) \right) \right\} \end{aligned}$$

which is exactly the same as $l_{Y|\mathbf{m}}(\boldsymbol{\beta} ; \mathbf{y})$.

Conclusively, if we let π_{jk} as above, then the log-linear model is the same as the logistic regression model for polytomous data.

Chapter 10. Nonlinear Regression Model

10.1 Nonlinear Regression Model

Let Y be a response and $\mathbf{x} = (X_1, X_2, \dots, X_k)'$ be k -dimensional covariate, then the regression model, in general, can be written

$$Y_i = f(\mathbf{x}_i, \boldsymbol{\theta}) + \varepsilon_i, \quad i = 1, 2, \dots, n$$

where $f(\mathbf{x}_i, \boldsymbol{\theta})$ is a regression function and $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)'$ is a p -dimensional parameter vector. So far, we assumed the multiple linear regression model, where $f(\mathbf{x}_i, \boldsymbol{\theta})$ is linear. Now, we assume that $f(\mathbf{x}_i, \boldsymbol{\theta})$ is nonlinear and the functional form is known.

For example, consider a first-order kinetics (material A transforms into material B) in chemistry, i.e.

$$\frac{\partial f_A(X, \theta)}{\partial X} = -\theta f_A(X, \theta), \quad \frac{\partial f_B(X, \theta)}{\partial X} = \theta f_A(X, \theta) \quad (\theta > 0)$$

where $f_A(X, \theta)$ and $f_B(X, \theta)$ denote density of A and B, respectively. Also, assume that at $X = 0$, the density of A is 100% and the density of B is zero.

Then, the solution of the above differential equation becomes

$$f_A(X, \theta) = e^{-\theta X}, \quad f_B(X, \theta) = 1 - e^{-\theta X}$$

Hence, the corresponding nonlinear regression model would be

$$Y_A = e^{-\theta X} + \varepsilon, \quad Y_B = 1 - e^{-\theta X} + \varepsilon$$

Ex.10.1(p.361)

Table 10.1 Heights of plant at given times (X : time, Y: height)

i	1	2	3	4	5	6	7	8	9
X_i	1	1.5	2	3	4	5	6	7.5	8.5
Y_i	2.1	2.5	4.9	5.5	7.0	8.4	9.6	10.2	11.4
i	10	11	12	13	14	15	16	17	18
X_i	10	12.5	15	17.5	20	25	30	35	40
Y_i	12.5	13.1	14.6	17.0	16.8	18.6	19.7	21.3	21.6

The popular nonlinear model for the growth (height) of a plant or an animal is given by

$$Y_i = \frac{\theta_1 X_i}{\theta_2 + X_i} + \varepsilon_i, \quad i = 1, 2, \dots, 18$$

Note that

$$X \rightarrow \infty \Rightarrow f(X, \theta_1, \theta_2) = \frac{\theta_1}{\theta_2 / X + 1} \rightarrow \theta_1$$

so that θ_1 can be regarded as the maximum height. Now, consider $X_{1/2}$, the time required to the half of the maximum height, then

$$\frac{1}{2}\theta_1 = \frac{\theta_1 X_{1/2}}{\theta_2 + X_{1/2}}$$

so that we have $\theta_2 = X_{1/2}$. In this data set, $\theta_1 \simeq 22$ and $\theta_2 \simeq 9$.

10.2 Estimation of Parameter

10.2.1 Method of least squares

To estimate θ , consider the method of least squares, i.e., minimize

$$S = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (Y_i - f(x_i, \theta))^2$$

Then,

$$\frac{\partial S}{\partial \theta_j} = (-2) \sum_{i=1}^n (Y_i - f(x_i, \theta)) \frac{\partial f(x_i, \theta)}{\partial \theta_j}, \quad j = 1, 2, \dots, p$$

and the estimate $\hat{\theta}$ satisfies

$$\sum_{i=1}^n f(x_i, \hat{\theta}) \frac{\partial f(x_i, \theta)}{\partial \theta_j} \Big|_{\theta=\hat{\theta}} = \sum_{i=1}^n Y_i \frac{\partial f(x_i, \theta)}{\partial \theta_j} \Big|_{\theta=\hat{\theta}}, \quad j = 1, 2, \dots, p$$

however, it is nonlinear in $\hat{\theta}$, and we cannot obtain the explicit solution,

i.e., we need iteration to get numerical value for $\hat{\theta}$. For example, consider

the first kinetics example,

$$Y_i = e^{-\theta X_i} + \varepsilon_i$$

and

$$f(X_i, \theta) = e^{-\theta X_i}, \quad \frac{\partial f(X_i, \theta)}{\partial \theta} = (-X_i) e^{-\theta X_i}$$

and therefore,

$$\sum_{i=1}^n X_i e^{-2\hat{\theta}X_i} = \sum_{i=1}^n X_i Y_i e^{-\hat{\theta}X_i}$$

which cannot be solved analytically.

10.2.2 Gauss-Newton method

The iteration method in the nonlinear regression is called the *Gauss – Newton* method. Let $\boldsymbol{\theta}^0 = (\theta_1^0, \theta_2^0, \dots, \theta_p^0)'$ be the initial vector for $\boldsymbol{\theta}$, then by the Taylor expansion, we have

$$f(x_i, \boldsymbol{\theta}) \approx f(x_i, \boldsymbol{\theta}^0) + \sum_{j=1}^p \frac{\partial f(x_i, \boldsymbol{\theta})}{\partial \theta_j} \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^0} (\theta_j - \theta_j^0)$$

and we replace it to the original model, then we have

$$Y_i = f(x_i, \boldsymbol{\theta}^0) + \sum_{j=1}^p \frac{\partial f(x_i, \boldsymbol{\theta})}{\partial \theta_j} \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^0} (\theta_j - \theta_j^0) + \varepsilon_i, \quad i = 1, 2, \dots, n$$

In matrix notation,

$$\mathbf{y} = \boldsymbol{\eta}(\boldsymbol{\theta}^0) + \mathbf{V}(\boldsymbol{\theta} - \boldsymbol{\theta}^0) + \boldsymbol{\varepsilon}$$

where

$$\boldsymbol{\eta}(\boldsymbol{\theta}^0) = (f(x_1, \boldsymbol{\theta}^0), \dots, f(x_n, \boldsymbol{\theta}^0))'$$

and V is $n \times p$ matrix with the (i, j) th component

$$v_{ij} = \frac{\partial f(x_i, \theta)}{\partial \theta_j} \Big|_{\theta=\theta^0}$$

and we call V *derivative matrix*. Note that the above equation can be written as

$$y - \eta(\theta^0) = V(\theta - \theta^0) + \varepsilon$$

then it is a form of linear regression model with the response $y^* = y - \eta(\theta^0)$ design matrix V , and parameter vector $\beta = (\theta - \theta^0)$, i.e.,

$$y^* = V\beta + \varepsilon$$

Hence, the estimator of β is

$$\hat{\beta} = (V'V)^{-1}V'y^* = (V'V)^{-1}V'(y - \eta(\theta^0))$$

which we let $\hat{\beta}^{(1)}$, and by $\beta = (\theta - \theta^0)$, we have

$$\hat{\theta}^{(1)} = \hat{\beta}^{(1)} + \theta^0$$

Now, we repeat this process, i.e.,

$$\hat{\theta}^{(i)} = \hat{\beta}^{(i)} + \hat{\theta}^{(i-1)}, \quad i = 1, 2, \dots$$

10.2.3 Computation of the initial value

Consider, for example, the first-order kinetics model

$$f_A(X_i, \theta) = e^{-\theta X_i}$$

If we take log on both sides, it becomes

$$\log f_A = -\theta X_i$$

which can be regarded as a simple linear regression without intercept.

Therefore, regress $\log Y_i$ on $(-X_i)$ to estimate the initial value for θ .

Ex.10.2(p.367)

10.3 Inference in the Nonlinear Model

Since the nonlinear model can be expressed as $\mathbf{y}^* = \mathbf{V}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, we can show that

$$\hat{\boldsymbol{\theta}} \rightarrow N(\boldsymbol{\theta}, (\hat{\mathbf{V}}' \hat{\mathbf{V}})^{-1} \sigma^2)$$

where $\hat{\mathbf{V}}$ is the estimator of $\mathbf{V}$. Also, the estimator of σ^2 is given by

$$s^2 = \frac{1}{n-p} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \frac{1}{n-p} \sum_{i=1}^n (Y_i - f(\mathbf{x}_i, \hat{\boldsymbol{\theta}}))^2$$

Using this result, the $100(1 - \alpha)\%$ simultaneous C.I. for $\boldsymbol{\theta}$ is obtained by solving

$$(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})' \hat{\mathbf{V}}' \hat{\mathbf{V}} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \leq p s^2 F_{\alpha}(p, n-p)$$

For the C.I. for θ_j , the standard error of $\hat{\theta}_j$ is

$$SE(\hat{\theta}_j) = \sqrt{c_{jj}} \cdot s$$

where c_{jj} is the j th diagonal element of $(\hat{\mathbf{V}}' \hat{\mathbf{V}})^{-1}$. Hence the $100(1 - \alpha)\%$

C.I for θ_j is

$$\hat{\theta}_j - t_{\alpha/2}(n - p)SE(\hat{\theta}_j) < \theta_j < \hat{\theta}_j + t_{\alpha/2}(n - p)SE(\hat{\theta}_j)$$

Now, for testing $H_0 : \theta_j = \theta_{j,0}$, the test statistic is

$$t_j = \frac{\hat{\theta}_j - \theta_{j,0}}{SE(\hat{\theta}_j)}$$

which is asymptotically $t(n - p)$. The hat matrix in the nonlinear model is

$$\mathbf{H} = \hat{\mathbf{V}}(\hat{\mathbf{V}}' \hat{\mathbf{V}})^{-1} \hat{\mathbf{V}}'$$

and therefore, the standardized residual becomes

$$r_i = \frac{Y_i - \hat{Y}_i}{s\sqrt{1 - h_{ii}}}$$

where h_{ii} is the i th diagonal element of $\mathbf{H}$.

Ex.10.3(p.370)